It is noted that some pneumatic relays are reverse acting. For example, the relay shown in Figure 5–28 is a reverse-acting relay. Here, as the nozzle back pressure P_b increases, the ball valve is forced toward the lower seat, thereby decreasing the control pressure P_c. Thus, this relay is a reverse-acting relay.

Pneumatic proportional controllers (force–distance type). Two types of pneumatic controllers, one called the force–distance type and the other the force–balance type, are used extensively in industry. Regardless of how differently industrial pneumatic controllers may appear, careful study will show the close similarity in the functions of the pneumatic circuit. Here we shall consider the force–distance type of pneumatic controllers.

Figure 5–29(a) shows a schematic diagram of such a proportional controller. The nozzle–flapper amplifier constitutes the first-stage amplifier, and the nozzle back pressure is controlled by the nozzle–flapper distance. The relay-type amplifier constitutes the second-stage amplifier. The nozzle back pressure determines the position of the diaphragm valve for the second-stage amplifier, which is capable of handling a large quantity of airflow.

In most pneumatic controllers, some type of pneumatic feedback is employed. Feedback of the pneumatic output reduces the amount of actual movement of the flapper. Instead of mounting the flapper on a fixed point, as shown in Figure 5–29(b), it is often pivoted on the feedback bellows, as shown in Figure 5–29(c). The amount of feedback can be regulated by introducing a variable linkage between the feedback bellows and the flapper connecting point. The flapper then becomes a floating link. It can be moved by both the error signal and the feedback signal.

The operation of the controller shown in Figure 5–29(a) is as follows. The input signal to the two-stage pneumatic amplifier is the actuating error signal. Increasing the actuating error signal moves the flapper to the left. This move will, in turn, increase the nozzle back pressure, and the diaphragm valve moves downward. This results in an increase of the control pressure. This increase will cause bellows F to expand and move the flapper to the right, thus opening the nozzle. Because of this feedback, the nozzle–flapper displacement is very small, but the change in the control pressure can be large.

It should be noted that proper operation of the controller requires that the feedback bellows move the flapper less than that movement caused by the error signal alone. (If these two movements were equal, no control action would result.)

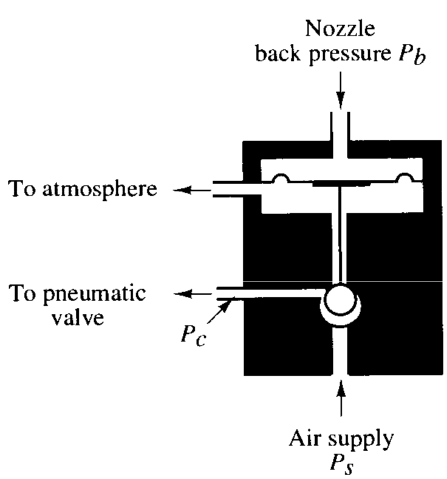


Figure 5–28
Reverse-acting relay.

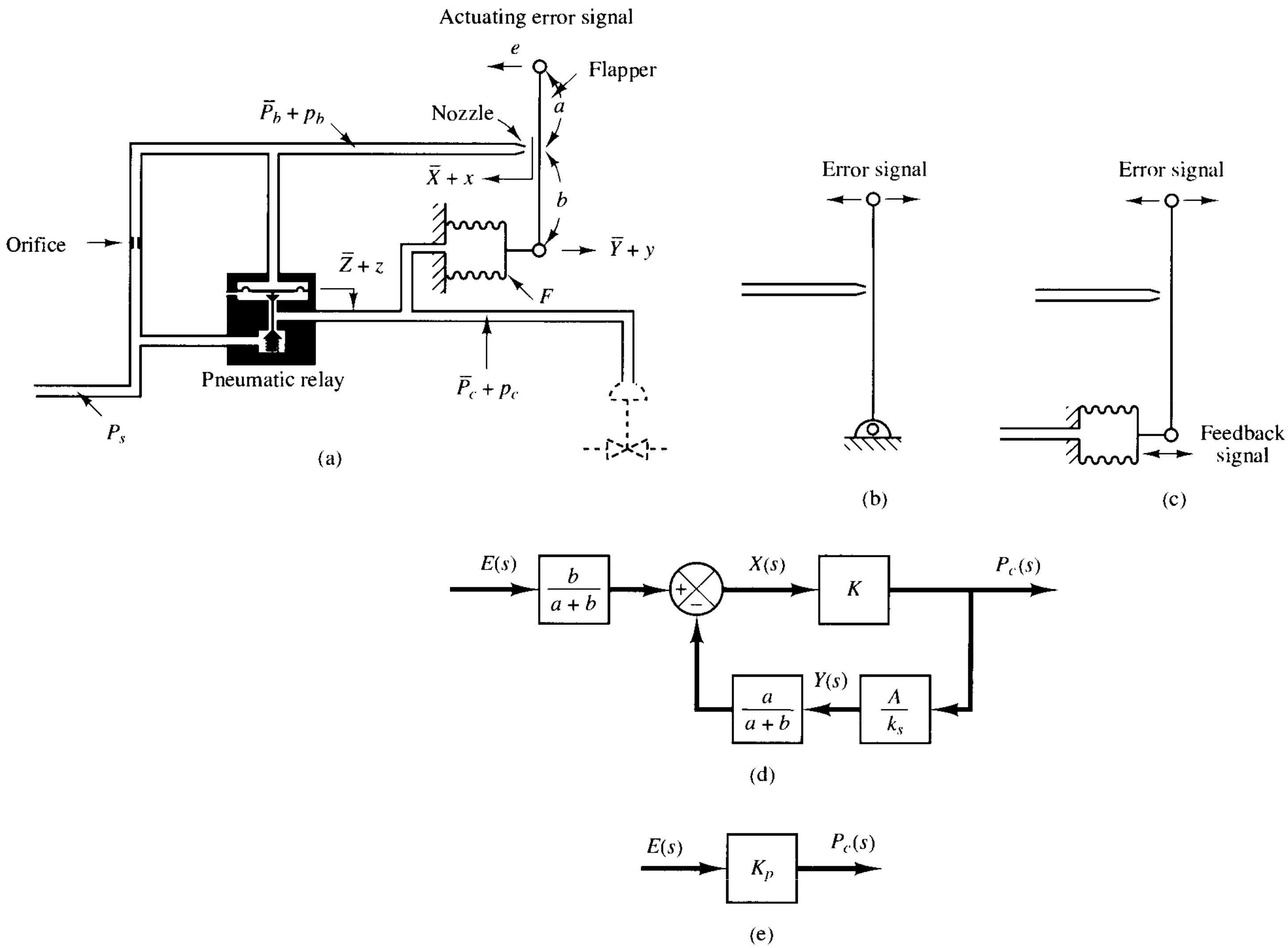


Figure 5–29
(a) Schematic diagram of a force–distance type of pneumatic proportional controller; (b) flapper mounted on a fixed point; (c) flapper mounted on a feedback bellows; (d) block diagram for the controller; (e) simplified block diagram for the controller.

Equations for this controller can be derived as follows. When the actuating error is zero, or $e = 0$, an equilibrium state exists with the nozzle–flapper distance equal to $\bar{X}$, the displacement of bellows equal to $\bar{Y}$, the displacement of the diaphragm equal to $\bar{Z}$, the nozzle back pressure equal to $\bar{P}_b$, and the control pressure equal to $\bar{P}_c$. When an actuating error exists, the nozzle–flapper distance, the displacement of the bellows, the displacement of the diaphragm, the nozzle back pressure, and the control pressure deviate from their respective equilibrium values. Let these deviations be x, y, z, p_b, and p_c, respectively. (The positive direction for each displacement variable is indicated by an arrowhead in the diagram.)

Assuming that the relationship between the variation in the nozzle back pressure and the variation in the nozzle–flapper distance is linear, we have

$$p_b = K_1 x \tag{5–13}$$

where K_1 is a positive constant. For the diaphragm valve,

$$p_b = K_2 z \tag{5–14}$$

where K_2 is a positive constant. The position of the diaphragm valve determines the control pressure. If the diaphragm valve is such that the relationship between p_c and z is linear, then

$$p_c = K_3 z \tag{5–15}$$

where K_3 is a positive constant. From Equations (5–13), (5–14), and (5–15), we obtain

$$p_c = \frac{K_3}{K_2} p_b = Kx \tag{5–16}$$

where $K = K_1 K_3 / K_2$ is a positive constant. For the flapper movement, we have

$$x = \frac{b}{a+b} e - \frac{a}{a+b} y \tag{5–17}$$

The bellows acts like a spring, and the following equation holds true:

$$A p_c = k_s y \tag{5–18}$$

where A is the effective area of the bellows and k_s is the equivalent spring constant, that is, the stiffness due to the action of the corrugated side of the bellows.

Assuming that all variations in the variables are within a linear range, we can obtain a block diagram for this system from Equations (5–16), (5–17), and (5–18) as shown in Figure 5–29(d). From Figure 5–29(d), it can be clearly seen that the pneumatic controller shown in Figure 5–29(a) itself is a feedback system. The transfer function between p_c and e is given by

$$\frac{P_c(s)}{E(s)} = \frac{\dfrac{b}{a+b} K}{1 + K \dfrac{a}{a+b} \dfrac{A}{k_s}} = K_p \tag{5–19}$$

A simplified block diagram is shown in Figure 5–29(e). Since p_c and e are proportional, the pneumatic controller shown in Figure 5–29(a) is called a *pneumatic proportional controller.* As seen from Equation (5–19), the gain of the pneumatic proportional controller can be widely varied by adjusting the flapper connecting linkage. [The flapper connecting linkage is not shown in Figure 5–29(a).] In most commercial proportional controllers an adjusting knob or other mechanism is provided for varying the gain by adjusting this linkage.

As noted earlier, the actuating error signal moved the flapper in one direction, and the feedback bellows moved the flapper in the opposite direction, but to a smaller degree. The effect of the feedback bellows is thus to reduce the sensitivity of the controller. The principle of feedback is commonly used to obtain wide proportional-band controllers.

Pneumatic controllers that do not have feedback mechanisms [which means that one end of the flapper is fixed, as shown in Figure 5–30(a)] have high sensitivity and are called *pneumatic two-position controllers* or *pneumatic on–off controllers*. In such a controller, only a small motion between the nozzle and the flapper is required to give a com-

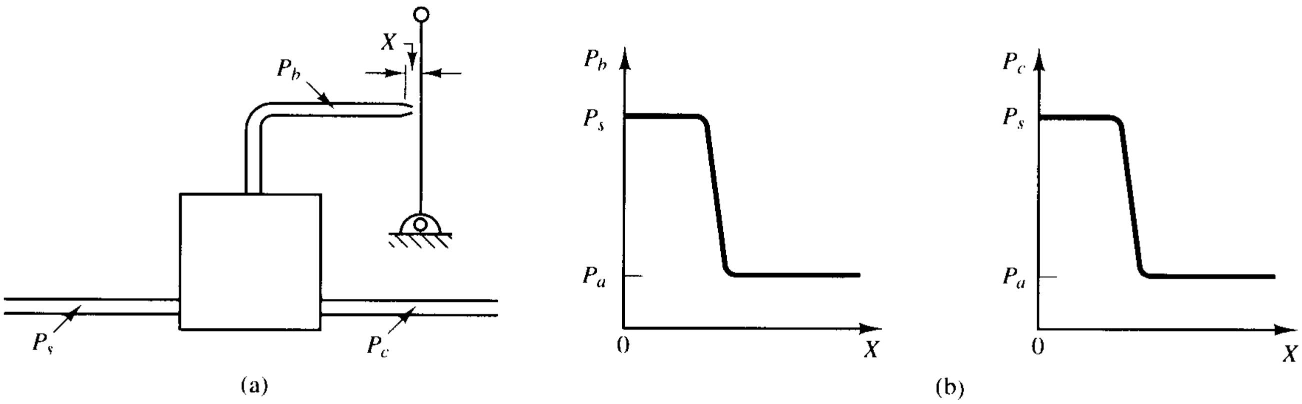


Figure 5–30
(a) Pneumatic controller without a feedback mechanism; (b) curves P_b versus X and P_c versus X.

plete change from the maximum to the minimum control pressure. The curves relating P_b to X and P_c to X are shown in Figure 5–30(b). Notice that a small change in X can cause a large change in P_b, which causes the diaphragm valve to be completely open or completely closed.

Pneumatic proportional controllers (force–balance type). Figure 5–31 shows a schematic diagram of a force–balance pneumatic proportional controller. Force–balance controllers are in extensive use in industry. Such controllers are called stack controllers. The basic principle of operation does not differ from that of the force–distance controller. The main advantage of the force–balance controller is that it eliminates many mechanical linkages and pivot joints, thereby reducing the effects of friction.

In what follows, we shall consider the principle of the force–balance controller. In the controller shown in Figure 5–31, the reference input pressure P_r and the output pressure P_o are fed to large diaphragm chambers. Note that a force-balance pneumatic controller operates only on pressure signals. Therefore, it is necessary to convert the reference input and system output to corresponding pressure signals.

As in the case of the force–distance controller, this controller employs a flapper, nozzle, and orifices. In Figure 5–31, the drilled opening in the bottom chamber is the nozzle. The diaphragm just above the nozzle acts as a flapper.

The operation of the force–balance controller shown in Figure 5–31 may be summarized as follows: 20-psig air from an air supply flows through an orifice, causing a reduced

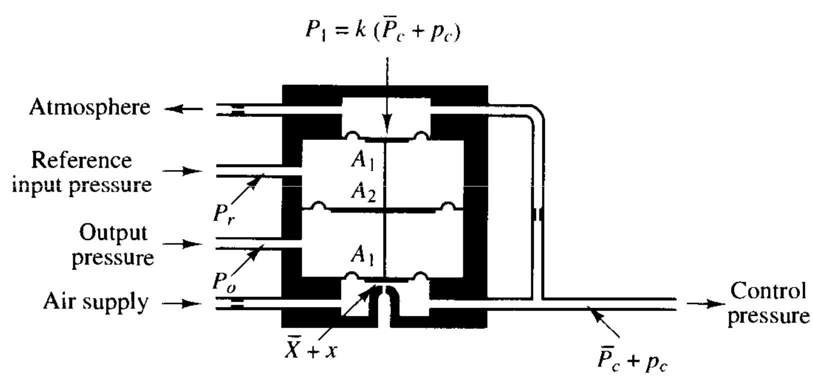


Figure 5–31
Schematic diagram of a force–balance type of pneumatic proportional controller.

pressure in the bottom chamber. Air in this chamber escapes to the atmosphere through the nozzle. The flow through the nozzle depends on the gap and the pressure drop across it. An increase in the reference input pressure P_r, while the output pressure P_o remains the same, causes the valve stem to move down, decreasing the gap between the nozzle and the flapper diaphragm. This causes the control pressure P_c to increase. Let

$$p_e = P_r - P_o \tag{5–20}$$

If $p_c = 0$, there is an equilibrium state with the nozzle–flapper distance equal to $\bar{X}$ and the control pressure equal to $\bar{P}_c$. At this equilibrium state, $P_1 = \bar{P}_c k$ (where $k < 1$) and

$$\bar{X} = \alpha(\bar{P}_c A_1 - \bar{P}_c k A_1) \tag{5–21}$$

where α is a constant.

Let us assume that $p_e \neq 0$ and define small variations in the nozzle–flapper distance and control pressure as x and p_c, respectively. Then we obtain the following equation:

$$\bar{X} + x = \alpha[(\bar{P}_c + p_c)A_1 - (\bar{P}_c + p_c)kA_1 - p_e(A_2 - A_1)] \tag{5–22}$$

From Equations (5–21) and (5–22), we obtain

$$x = \alpha[p_c(1 - k)A_1 - p_e(A_2 - A_1)] \tag{5–23}$$

At this point, we must examine the quantity x. In the design of pneumatic controllers, the nozzle–flapper distance is made quite small. In view of the fact that x/α is a higher-order term than $p_c(1 - k)A_1$ or $p_e(A_2 - A_1)$, that is, for $p_c \neq 0$,

$$\frac{x}{\alpha} \ll p_c(1 - k)A_1$$

$$\frac{x}{\alpha} \ll p_e(A_2 - A_1)$$

we may neglect the term x in our analysis. Equation (5–23) can then be rewritten to reflect this assumption as follows:

$$p_c(1 - k)A_1 = p_e(A_2 - A_1)$$

and the transfer function between p_c and p_e becomes

$$\frac{P_c(s)}{P_e(s)} = \frac{A_2 - A_1}{A_1} \frac{1}{1 - k} = K_p$$

where p_e is defined by Equation (5–20). The controller shown in Figure 5–31 is a proportional controller. The value of gain K_p increases as k approaches unity. Note that the value of k depends on the diameters of the orifices in the inlet and outlet pipes of the feedback chamber. (The value of k approaches unity as the resistance to flow in the orifice of the inlet pipe is made smaller.)

Pneumatic actuating valves. One characteristic of pneumatic controls is that they almost exclusively employ pneumatic actuating valves. A pneumatic actuating valve can provide a large power output. (Since a pneumatic actuator requires a large power input to produce a large power output, it is necessary that a sufficient quantity of pressurized air be available.) In practical pneumatic actuating valves, the valve characteristics may not be linear; that is, the flow may not be directly proportional to the valve stem position, and also there may be other nonlinear effects, such as hysteresis.

Consider the schematic diagram of a pneumatic actuating valve shown in Figure 5–32. Assume that the area of the diaphragm is A. Assume also that when the actuating error is zero the control pressure is equal to $\bar{P}_c$ and the valve displacement is equal to $\bar{X}$.

In the following analysis, we shall consider small variations in the variables and linearize the pneumatic actuating valve. Let us define the small variation in the control pressure and the corresponding valve displacement to be p_c and x, respectively. Since a small change in the pneumatic pressure force applied to the diaphragm repositions the load, consisting of the spring, viscous friction, and mass, the force balance equation becomes

$$Ap_c = m\ddot{x} + b\dot{x} + kx \tag{5–24}$$

where m = mass of the valve and valve stem
b = viscous-friction coefficient
k = spring constant

If the force due to the mass and viscous friction are negligibly small, then Equation (5–24) can be simplified to:

$$Ap_c = kx$$

The transfer function between x and p_c thus becomes

$$\frac{X(s)}{P_c(s)} = \frac{A}{k} = K_c$$

where $X(s) = \mathcal{L}[x]$ and $P_c(s) = \mathcal{L}[p_c]$. If q_i, the change in flow through the pneumatic actuating valve, is proportional to x, the change in the valve-stem displacement, then

$$\frac{Q_i(s)}{X(s)} = K_q$$

where $Q_i(s) = \mathcal{L}[q_i]$ and K_q is a constant. The transfer function between q_i and p_c becomes

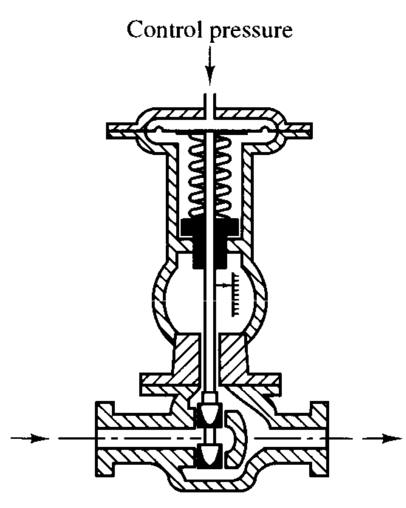


Figure 5–32
Schematic diagram of a pneumatic actuating valve.

$$\frac{Q_i(s)}{P_c(s)} = K_c K_q = K_v$$

where K_v is a constant.

The standard control pressure for this kind of a pneumatic actuating valve is between 3 and 15 psig. The valve-stem displacement is limited by the allowable stroke of the diaphragm and is only a few inches. If a longer stroke is needed, a piston–spring combination may be employed.

In pneumatic actuating valves, the static-friction force must be limited to a low value so that excessive hysteresis does not result. Because of the compressibility of air, the control action may not be positive; that is, an error may exist in the valve-stem position. The use of a valve positioner results in improvements in the performance of a pneumatic actuating valve.

Basic principle for obtaining derivative control action. We shall now present methods for obtaining derivative control action. We shall again place the emphasis on the principle and not on the details of the actual mechanisms.

The basic principle for generating a desired control action is to insert the inverse of the desired transfer function in the feedback path. For the system shown in Figure 5–33, the closed-loop transfer function is

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}$$

If $|G(s)H(s)| \gg 1$, then $C(s)/R(s)$ can be modified to

$$\frac{C(s)}{R(s)} = \frac{1}{H(s)}$$

Thus, if proportional-plus-derivative control action is desired, we insert an element having the transfer function $1/(Ts + 1)$ in the feedback path.

Consider the pneumatic controller shown in Figure 5–34(a). Considering small changes in the variables, we can draw a block diagram of this controller as shown in Figure 5–34(b). From the block diagram we see that the controller is of proportional type.

We shall now show that the addition of a restriction in the negative feedback path will modify the proportional controller to a proportional-plus-derivative controller, commonly called a PD controller.

Consider the pneumatic controller shown in Figure 5–35(a). Assuming again small changes in the actuating error, nozzle–flapper distance, and control pressure, we can

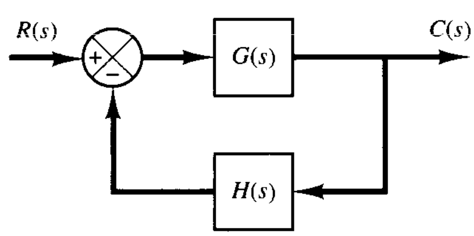


Figure 5–33
Control system.

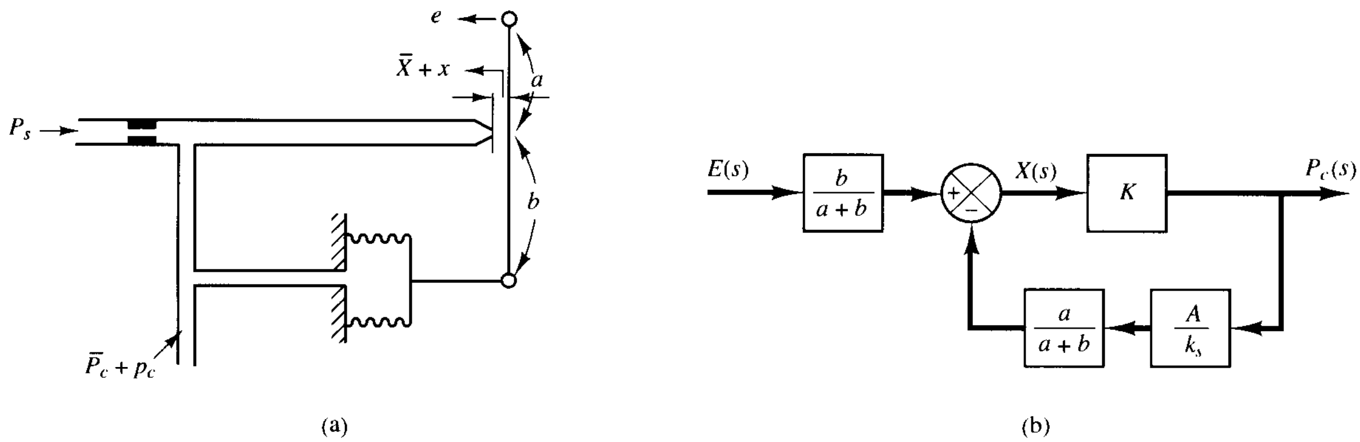


Figure 5–34
(a) Pneumatic proportional controller; (b) block diagram of the controller.

summarize the operation of this controller as follows: Let us first assume a small step change in e. Then the change in the control pressure p_c will be instantaneous. The restriction R will momentarily prevent the feedback bellows from sensing the pressure change p_c. Thus the feedback bellows will not respond momentarily, and the pneumatic actuating valve will feel the full effect of the movement of the flapper. As time goes on, the feedback bellows will expand or contract. The change in the nozzle-flapper distance x and the change in the control pressure p_c can be plotted against time t, as shown in Figure 5–35(b). At steady state, the feedback bellows acts like an ordinary feedback

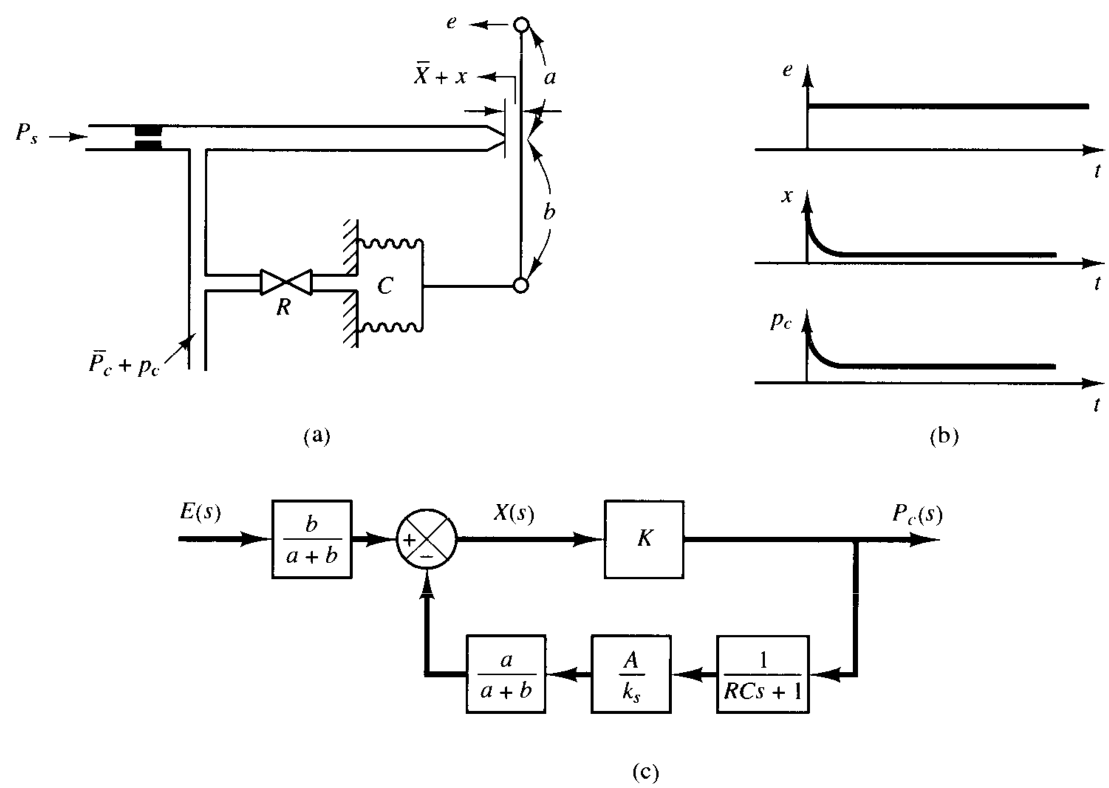


Figure 5–35
(a) Pneumatic proportional-plus-derivative controller; (b) step change in e and the corresponding changes in x and p_c plotted versus t; (c) block diagram of the controller.

mechanism. The curve p_c versus t clearly shows that this controller is of the proportional-plus-derivative type.

A block diagram corresponding to this pneumatic controller is shown in Figure 5–35(c). In the block diagram, K is a constant, A is the area of the bellows, and k_s is the equivalent spring constant of the bellows. The transfer function between p_c and e can be obtained from the block diagram as follows:

$$\frac{P_c(s)}{E(s)} = \frac{\dfrac{b}{a + b}K}{1 + \dfrac{Ka}{a + b}\dfrac{A}{k_s}\dfrac{1}{RCs + 1}}$$

In such a controller the loop gain $|KaA/[(a + b)k_s(RCs + 1)]|$ is normally very much greater than unity. Thus the transfer function $P_c(s)/E(s)$ can be simplified to give

$$\frac{P_c(s)}{E(s)} = K_p(1 + T_d s)$$

where

$$K_p = \frac{bk_s}{aA}, \qquad T_d = RC$$

Thus, delayed negative feedback, or the transfer function $1/(RCs + 1)$ in the feedback path, modifies the proportional controller to a proportional-plus-derivative controller.

Note that if the feedback valve is fully opened the control action becomes proportional. If the feedback valve is fully closed, the control action becomes narrow-band proportional (on–off).

Obtaining pneumatic proportional-plus-integral control action. Consider the proportional controller shown in Figure 5–34(a). Considering small changes in the variables, we can show that the addition of delayed positive feedback will modify this proportional controller to a proportional-plus-integral controller, commonly called a PI controller.

Consider the pneumatic controller shown in Figure 5–36(a). The operation of this controller is as follows: The bellows denoted by I is connected to the control pressure source without any restriction. The bellows denoted by II is connected to the control pressure source through a restriction. Let us assume a small step change in the actuating error. This will cause the back pressure in the nozzle to change instantaneously. Thus a change in the control pressure p_c also occurs instantaneously. Due to the restriction of the valve in the path to bellows II, there will be a pressure drop across the valve. As time goes on, air will flow across the valve in such a way that the change in pressure in bellows II attains the value p_c. Thus bellows II will expand or contract as time elapses in such a way as to move the flapper an additional amount in the direction of the original

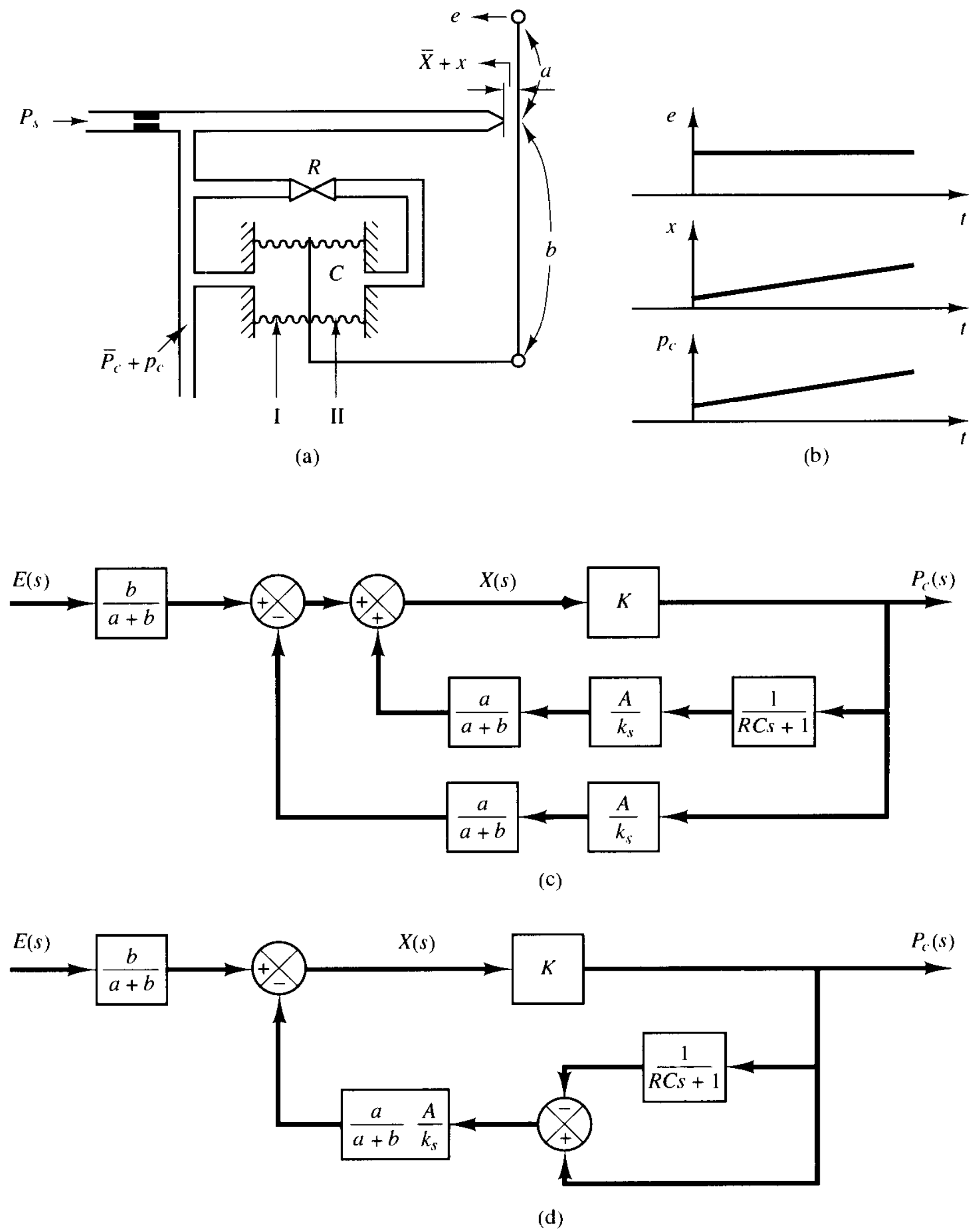


Figure 5–36
(a) Pneumatic proportional-plus-integral controller; (b) step change in e and the corresponding changes in x and p_c plotted versus t; (c) block diagram of the controller; (d) simplified block diagram.

displacement e. This will cause the back pressure p_c in the nozzle to change continuously, as shown in Figure 5–36(b).

Note that the integral control action in the controller takes the form of slowly canceling the feedback that the proportional control originally provided.

A block diagram of this controller under the assumption of small variations in the variables is shown in Figure 5–36(c). A simplification of this block diagram yields Figure 5–36(d). The transfer function of this controller is

$$\frac{P_c(s)}{E(s)} = \frac{\dfrac{b}{a+b}K}{1 + \dfrac{Ka}{a+b}\dfrac{A}{k_s}\left(1 - \dfrac{1}{RCs+1}\right)}$$

where K is a constant, A is the area of the bellows, and k_s is the equivalent spring constant of the combined bellows. If $|KaARCs/[(a+b)k_s(RCs+1)]| \gg 1$, which is usually the case, the transfer function can be simplified to

$$\frac{P_c(s)}{E(s)} = K_p\left(1 + \frac{1}{T_i s}\right)$$

where

$$K_p = \frac{bk_s}{aA}, \qquad T_i = RC$$

Obtaining pneumatic proportional-plus-integral-plus-derivative control action. A combination of the pneumatic controllers shown in Figures 5–35(a) and 5–36(a) yields a proportional-plus-integral-plus-derivative controller, commonly called a PID controller. Figure 5–37(a) shows a schematic diagram of such a controller. Figure 5–37(b) shows a block diagram of this controller under the assumption of small variations in the variables.

The transfer function of this controller is

$$\frac{P_c(s)}{E(s)} = \frac{\dfrac{bK}{a+b}}{1 + \dfrac{Ka}{a+b}\dfrac{A}{k_s}\dfrac{(R_iC - R_dC)s}{(R_dCs+1)(R_iCs+1)}}$$

By defining

$$T_i = R_iC, \qquad T_d = R_dC$$

and noting that under normal operation $|KaA(T_i - T_d)s/[(a+b)k_s(T_ds+1)(T_is+1)]| \gg 1$ and $T_i \gg T_d$, we obtain

$$\begin{aligned}\frac{P_c(s)}{E(s)} &\doteq \frac{bk_s}{aA}\frac{(T_ds+1)(T_is+1)}{(T_i - T_d)s} \\ &\doteq \frac{bk_s}{aA}\frac{T_dT_is^2 + T_is + 1}{T_is} \\ &= K_p\left(1 + \frac{1}{T_is} + T_ds\right)\end{aligned} \tag{5–25}$$

where

$$K_p = \frac{bk_s}{aA}$$

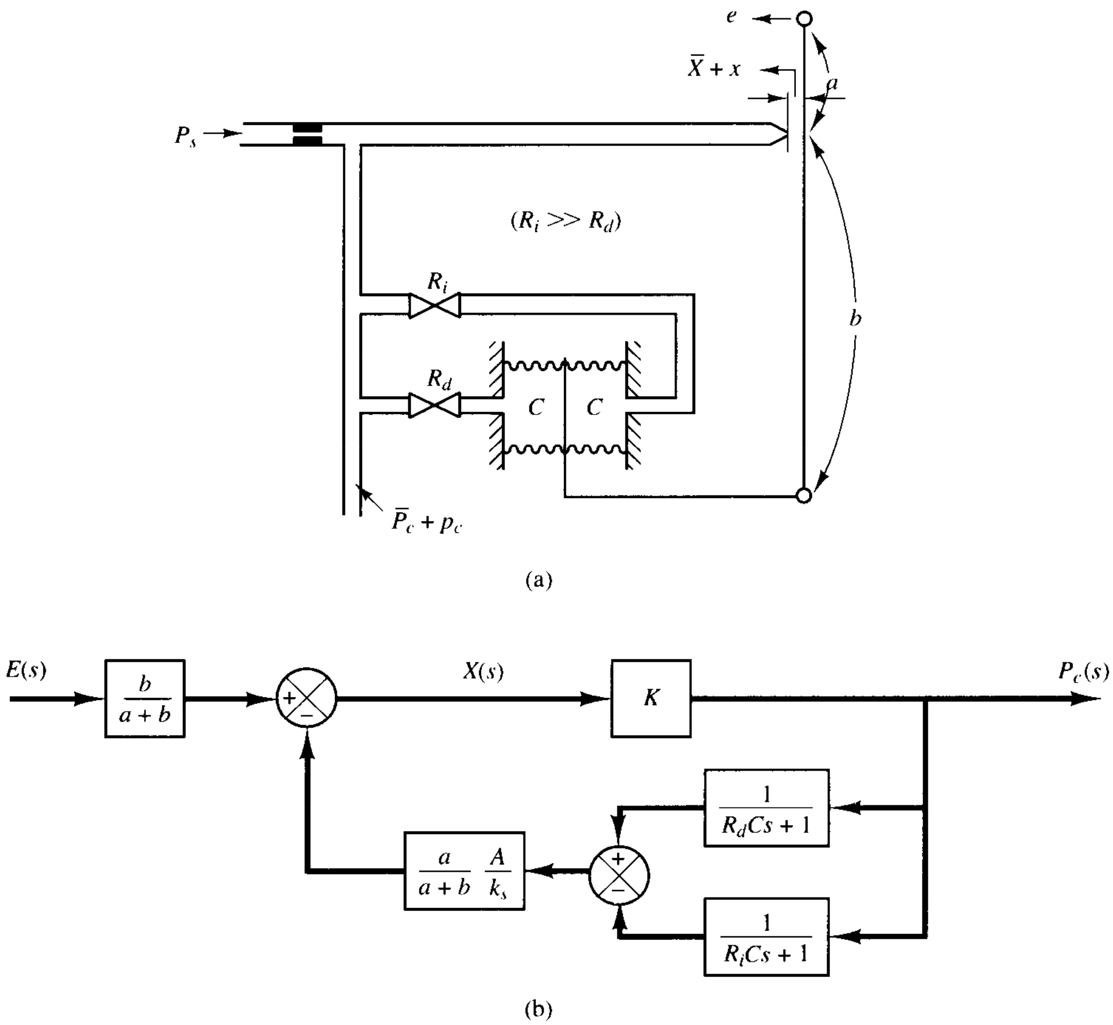


Figure 5–37
(a) Pneumatic proportional-plus-integral-plus-derivative controller; (b) block diagram of the controller.

Equation (5–25) indicates that the controller shown in Figure 5–37(a) is a proportional-plus-integral-plus-derivative controller (a PID controller).

5–7 HYDRAULIC CONTROLLERS

Except for low-pressure pneumatic controllers, compressed air has seldom been used for the continuous control of the motion of devices having significant mass under external load forces. For such a case, hydraulic controllers are generally preferred.

Hydraulic systems. The widespread use of hydraulic circuitry in machine tool applications, aircraft control systems, and similar operations occurs because of such factors as positiveness, accuracy, flexibility, high horsepower-to-weight ratio, fast starting, stopping, and reversal with smoothness and precision, and simplicity of operations.

The operating pressure in hydraulic systems is somewhere between 145 and 5000 $\text{lb}_f/\text{in.}^2$ (between 1 and 35 MPa). In some special applications, the operating pressure may go up to 10,000 $\text{lb}_f/\text{in.}^2$ (70 MPa). For the same power requirement, the weight and size of the hydraulic unit can be made smaller by increasing the supply pressure. With high-pressure hydraulic systems, very large force can be obtained. Rapid-acting, accurate positioning of heavy loads is possible with hydraulic systems. A combination of electronic and hydraulic systems is widely used because it combines the advantages of both electronic control and hydraulic power.

Advantages and disadvantages of hydraulic systems. There are certain advantages and disadvantages in using hydraulic systems rather than other systems. Some of the advantages are the following:

1. Hydraulic fluid acts as a lubricant, in addition to carrying away heat generated in the system to a convenient heat exchanger.
2. Comparatively small sized hydraulic actuators can develop large forces or torques.
3. Hydraulic actuators have a higher speed of response with fast starts, stops, and speed reversals.
4. Hydraulic actuators can be operated under continuous, intermittent, reversing, and stalled conditions without damage.
5. Availability of both linear and rotary actuators gives flexibility in design.
6. Because of low leakages in hydraulic actuators, speed drop when loads are applied is small.

On the other hand, several disadvantages tend to limit their use.

1. Hydraulic power is not readily available compared to electric power.
2. Cost of a hydraulic system may be higher than a comparable electrical system performing a similar function.
3. Fire and explosion hazards exist unless fire-resistant fluids are used.
4. Because it is difficult to maintain a hydraulic system that is free from leaks, the system tends to be messy.
5. Contaminated oil may cause failure in the proper functioning of a hydraulic system.
6. As a result of the nonlinear and other complex characteristics involved, the design of sophisticated hydraulic systems is quite involved.
7. Hydraulic circuits have generally poor damping characteristics. If a hydraulic circuit is not designed properly, some unstable phenomena may occur or disappear, depending on the operating condition.

Comments. Particular attention is necessary to ensure that the hydraulic system is stable and satisfactory under all operating conditions. Since the viscosity of hydraulic fluid can greatly affect damping and friction effects of the hydraulic circuits, stability tests must be carried out at the highest possible operating temperature.

Note that most hydraulic systems are nonlinear. Sometimes, however, it is possible to linearize nonlinear systems so as to reduce their complexity and permit solutions that are sufficiently accurate for most purposes. A useful linearization technique for dealing with nonlinear systems was presented in Section 3–10.

Hydraulic integral controllers. The hydraulic servomotor shown in Figure 5–38 is essentially a pilot-valve-controlled hydraulic power amplifier and actuator. The pilot valve is a balanced valve in the sense that the pressure forces acting on it are all bal-

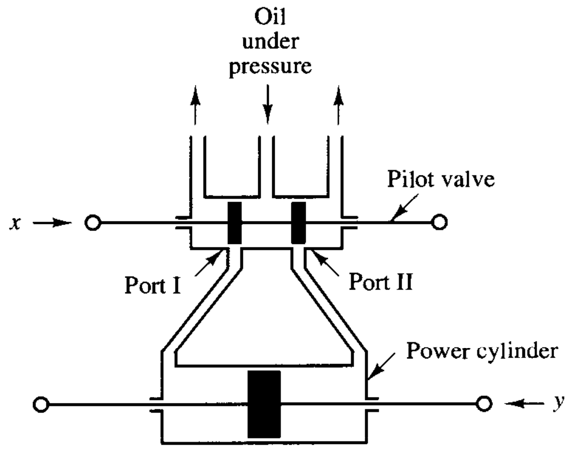


Figure 5–38
Hydraulic servomotor.

anced. A very large power output can be controlled by a pilot valve, which can be positioned with very little power.

It will be shown in the following that for negligibly small load mass the servomotor shown in Figure 5–38 acts as an integrator or an integral controller. Such a servomotor constitutes the basis of the hydraulic control circuit.

In the hydraulic servomotor shown in Figure 5–38, the pilot valve (a four-way valve) has two lands on the spool. If the width of the land is smaller than the port in the valve sleeve, the valve is said to be *underlapped. Overlapped* valves have a land width greater than the port width. A *zero-lapped* valve has a land width that is identical to the port width. (If the pilot valve is not a zero-lapped valve, analyses of hydraulic servomotors become very complicated.)

In the present analysis, we assume that hydraulic fluid is incompressible and that the inertia force of the power piston and load is negligible compared to the hydraulic force at the power piston. We also assume that the pilot valve is a zero-lapped valve, and the oil flow rate is proportional to the pilot valve displacement.

Operation of this hydraulic servomotor is as follows. If input x moves the pilot valve to the right, port II is uncovered, and so high-pressure oil enters the right side of the power piston. Since port I is connected to the drain port, the oil in the left side of the power piston is returned to the drain. The oil flowing into the power cylinder is at high pressure; the oil flowing out from the power cylinder into the drain is at low pressure. The resulting difference in pressure on both sides of the power piston will cause it to move to the left.

Note that the rate of flow of oil q (kg/sec) times dt (sec) is equal to the power piston displacement dy(m) times the piston area A(m^2) times the density of oil ρ (kg/m^3). Therefore,

$$A\rho\, dy = q\, dt \tag{5–26}$$

Because of the assumption that the oil flow rate q is proportional to the pilot valve displacement x, we have

$$q = K_1 x \tag{5–27}$$

where K_1 is a positive constant. From Equations (5–26) and (5–27) we obtain

$$A\rho \frac{dy}{dt} = K_1 x$$

The Laplace transform of this last equation, assuming a zero initial condition, gives

$$A\rho s Y(s) = K_1 X(s)$$

or

$$\frac{Y(s)}{X(s)} = \frac{K_1}{A\rho s} = \frac{K}{s}$$

where $K = K_1/(A\rho)$. Thus the hydraulic servomotor shown in Figure 5–38 acts as an integral controller.

Hydraulic proportional controllers. It has been shown that the servomotor in Figure 5–38 acts as an integral controller. This servomotor can be modified to a proportional controller by means of a feedback link. Consider the hydraulic controller shown in Figure 5–39(a). The left side of the pilot valve is joined to the left side of the power piston by a link ABC. This link is a floating link rather than one moving about a fixed pivot.

The controller here operates in the following way. If input e moves the pilot valve to the right, port II will be uncovered and high-pressure oil will flow through port II into the right side of the power piston and force this piston to the left. The power piston, in moving to the left, will carry the feedback link ABC with it, thereby moving the pilot valve to the left. This action continues until the pilot piston again covers ports I and II. A block diagram of the system can be drawn as in Figure 5–39(b). The transfer function between $Y(s)$ and $E(s)$ is given by

$$\frac{Y(s)}{E(s)} = \frac{\dfrac{b}{a+b}\dfrac{K}{s}}{1 + \dfrac{K}{s}\dfrac{a}{a+b}}$$

$$= \frac{bK}{s(a+b) + Ka}$$

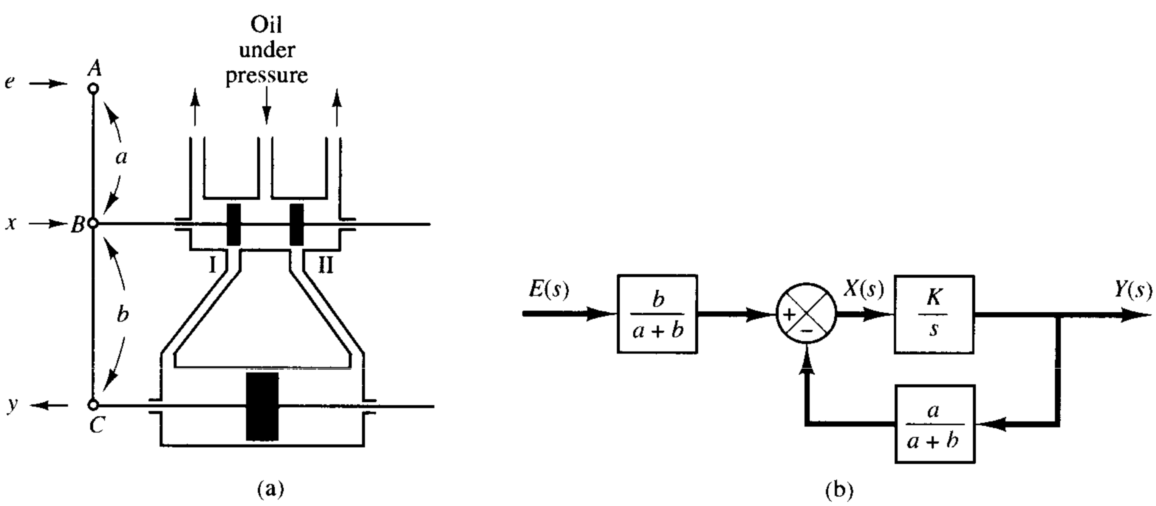


Figure 5–39
(a) Servomotor that acts as a proportional controller; (b) block diagram of the servomotor.

Noting that under the normal operating conditions we have $|Ka/[s(a+b)]| \gg 1$, this last equation can be simplified to

$$\frac{Y(s)}{E(s)} = \frac{b}{a} = K_p$$

The transfer function between y and e becomes a constant. Thus, the hydraulic controller shown in Figure 5–39(a) acts as a proportional controller, the gain of which is K_p. This gain can be adjusted by effectively changing the lever ratio b/a. (The adjusting mechanism is not shown in the diagram.)

We have thus seen that the addition of a feedback link will cause the hydraulic servomotor to act as a proportional controller.

Dashpots. The dashpot (also called a damper) shown in Figure 5–40(a) acts as a differentiating element. Suppose that we introduce a step displacement to the piston position x. Then the displacement y becomes equal to x momentarily. Because of the spring force, however, the oil will flow through the resistance R and the cylinder will come back to the original position. The curves x versus t and y versus t are shown in Figure 5–40(b).

Let us derive the transfer function between the displacement y and displacement x. Define the pressures existing on the right and left sides of the piston as P_1 (lb/in.^{2}) and P_2 (lb/in.^{2}), respectively. Suppose that the inertia force involved is negligible. Then the force acting on the piston must balance the spring force. Thus

$$A(P_1 - P_2) = ky$$

where A = piston area, in.^{2}
k = spring constant, lb$_f$/in.

The flow rate q is given by

$$q = \frac{P_1 - P_2}{R}$$

where q = flow rate through the restriction, lb/sec
R = resistance to flow at the restriction, lb$_f$-sec/in.^{2}-lb

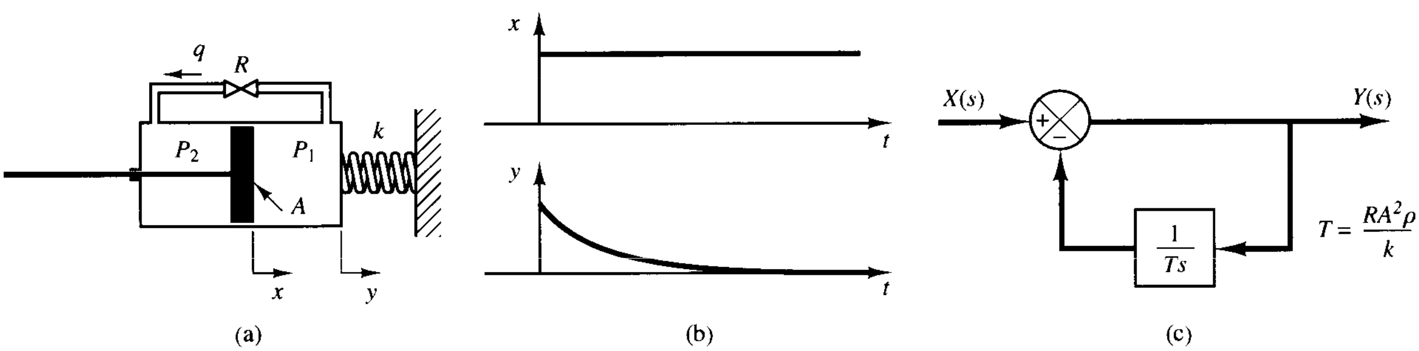


Figure 5–40
(a) Dashpot; (b) step change in x and the corresponding change in y plotted versus t; (c) block diagram of the dashpot.

Since the flow through the restriction during dt seconds must equal the change in the mass of oil to the left of the piston during the same dt seconds, we obtain

$$q\,dt = A\rho(dx - dy)$$

where ρ = density, lb/in.^{3}. (We assume that the fluid is incompressible or ρ = constant.) This last equation can be rewritten as

$$\frac{dx}{dt} - \frac{dy}{dt} = \frac{q}{A\rho} = \frac{P_1 - P_2}{RA\rho} = \frac{ky}{RA^2\rho}$$

or

$$\frac{dx}{dt} = \frac{dy}{dt} + \frac{ky}{RA^2\rho}$$

Taking the Laplace transforms of both sides of this last equation, assuming zero initial conditions, we obtain

$$sX(s) = sY(s) + \frac{k}{RA^2\rho}Y(s)$$

The transfer function of this system thus becomes

$$\frac{Y(s)}{X(s)} = \frac{s}{s + \dfrac{k}{RA^2\rho}}$$

Let us define $RA^2\rho/k = T$. Then

$$\frac{Y(s)}{X(s)} = \frac{Ts}{Ts + 1} = \frac{1}{1 + \dfrac{1}{Ts}}$$

Figure 5–40(c) shows a block diagram representation for this system.

Obtaining hydraulic proportional-plus-integral control action. Figure 5–41(a) shows a schematic diagram of a hydraulic proportional-plus-integral controller. A block diagram of this controller is shown in Figure 5–41(b). The transfer function $Y(s)/E(s)$ is given by

$$\frac{Y(s)}{E(s)} = \frac{\dfrac{b}{a+b}\dfrac{K}{s}}{1 + \dfrac{Ka}{a+b}\dfrac{T}{Ts+1}}$$

In such a controller, under normal operation $|KaT/[(a + b)(Ts + 1)]| \gg 1$, with the result that

$$\frac{Y(s)}{E(s)} = K_p\left(1 + \frac{1}{T_i s}\right)$$

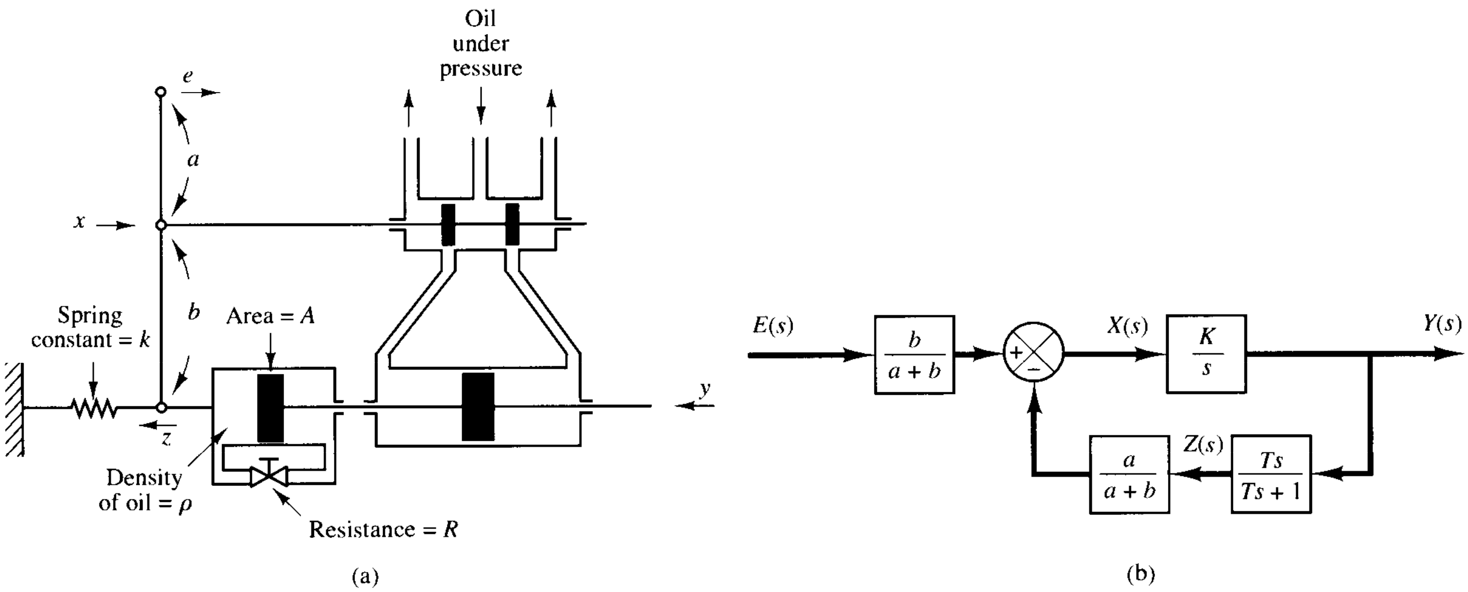


Figure 5–41
(a) Schematic diagram of a hydraulic proportional-plus-integral controller; (b) block diagram of the controller.

where

$$K_p = \frac{b}{a}, \qquad T_i = T = \frac{RA^2\rho}{k}$$

Thus the controller shown in Figure 5–52(a) is a proportional-plus-integral controller (a PI controller.)

Obtaining hydraulic proportional-plus-derivative control action. Figure 5–42(a) shows a schematic diagram of a hydraulic proportional-plus-derivative

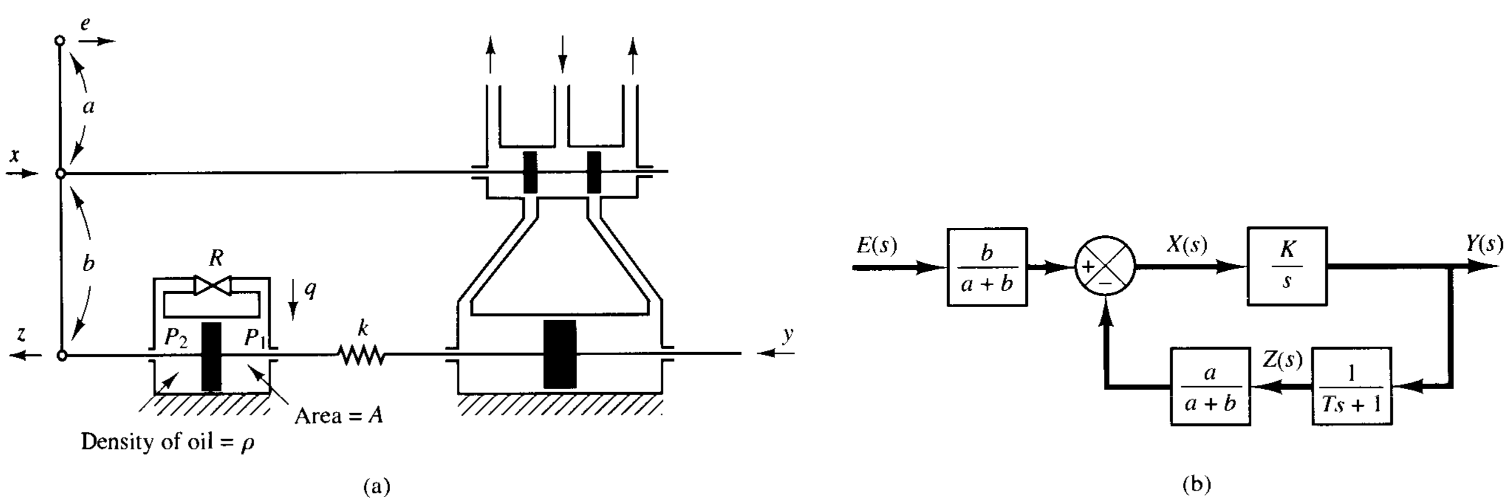


Figure 5–42
(a) Schematic diagram of a hydraulic proportional-plus-derivative controller; (b) block diagram of the controller.

controller. The cylinders are fixed in space and the pistons can move. For this system, notice that

$$k(y - z) = A(P_2 - P_1)$$

$$q = \frac{P_2 - P_1}{R}$$

$$q\,dt = \rho A\,dz$$

Hence

$$y = z + \frac{A}{k} qR = z + \frac{RA^2\rho}{k}\frac{dz}{dt}$$

or

$$\frac{Z(s)}{Y(s)} = \frac{1}{Ts + 1}$$

where

$$T = \frac{RA^2\rho}{k}$$

A block diagram for this system is shown in Figure 5–42(b). From the block diagram the transfer function $Y(s)/E(s)$ can be obtained as

$$\frac{Y(s)}{E(s)} = \frac{\dfrac{b}{a + b}\dfrac{K}{s}}{1 + \dfrac{a}{a + b}\dfrac{K}{s}\dfrac{1}{Ts + 1}}$$

Under normal operation we have $|aK/[(a + b)s(Ts + 1)]| \gg 1$. Hence

$$\frac{Y(s)}{E(s)} = K_p(1 + Ts)$$

where

$$K_p = \frac{b}{a}, \qquad T = \frac{RA^2\rho}{k}$$

Thus the controller shown in Figure 5–42(a) is a proportional-plus-derivative controller (a PD controller).

5–8 ELECTRONIC CONTROLLERS

This section discusses electronic controllers using operational amplifiers. We begin by deriving the transfer functions of simple operational-amplifier circuits. Then we derive the transfer functions of some of the operational-amplifier controllers. Finally, we give operational-amplifier controllers and their transfer functions in the form of a table.

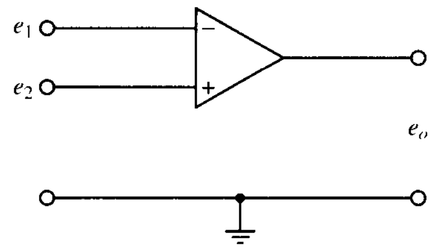


Figure 5–43
Operational amplifier.

Operational amplifiers. Operational amplifiers, often called op amps, are frequently used to amplify signals in sensor circuits. Op amps are also frequently used in filters used for compensation purposes. Figure 5–43 shows an op amp. It is a common practice to choose the ground as 0 volt and measure the input voltages e_1 and e_2 relative to the ground. The input e_1 to the minus terminal of the amplifier is inverted, and the input e_2 to the plus terminal is not inverted. The total input to the amplifier thus becomes $e_2 - e_1$. Hence, for the circuit shown in Figure 5–54, we have

$$e_o = K(e_2 - e_1) = -K(e_1 - e_2)$$

where the inputs e_1 and e_2 may be dc or ac signals and K is the differential gain or voltage gain. The magnitude of K is approximately $10^5 \sim 10^6$ for dc signals and ac signals with frequencies less than approximately 10 Hz. (The differential gain K decreases with the signal frequency and becomes about unity for frequencies of 1 MHz ~ 50 MHz.) Note that the op amp amplifies the difference in voltages e_1 and e_2. Such an amplifier is commonly called a differential amplifier. Since the gain of the op amp is very high, it is necessary to have a negative feedback from the output to the input to make the amplifier stable. (The feedback is made from the output to the inverted input so that the feedback is a negative feedback.)

In the ideal op amp, no current flows into the input terminals, and the output voltage is not affected by the load connected to the output terminal. In other words, the input impedance is infinity and the output impedance is zero. In an actual op amp, a very small (almost negligible) current flows into an input terminal and the output cannot be loaded too much. In our analysis here, we make the assumption that the op amps are ideal.

Inverting amplifier. Consider the operational amplifier circuit shown in Figure 5–44. Let us obtain the output voltage e_o.

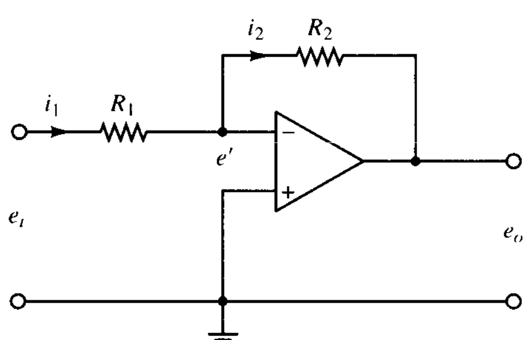


Figure 5–44
Inverting amplifier.

The equation for this circuit can be obtained as follows: Define

$$i_1 = \frac{e_i - e'}{R_1}, \qquad i_2 = \frac{e' - e_o}{R_2}$$

Since only a negligible current flows into the amplifier, the current i_1 must be equal to current i_2. Thus

$$\frac{e_i - e'}{R_1} = \frac{e' - e_o}{R_2}$$

Since $K(0 - e') = e_0$ and $K \gg 1$, e' must be almost zero, or $e' \doteqdot 0$. Hence we have

$$\frac{e_i}{R_1} = \frac{-e_o}{R_2}$$

or

$$e_o = -\frac{R_2}{R_1} e_i$$

Thus the circuit shown is an inverting amplifier. If $R_1 = R_2$, then the op-amp circuit shown acts as a sign inverter.

Noninverting amplifier. Figure 5–45(a) shows a noninverting amplifier. A circuit equivalent to this one is shown in Figure 5–45(b). For the circuit of Figure 5–45(b), we have

$$e_o = K\left(e_i - \frac{R_1}{R_1 + R_2} e_o\right)$$

where K is the differential gain of the amplifier. From this last equation, we get

$$e_i = \left(\frac{R_1}{R_1 + R_2} + \frac{1}{K}\right) e_o$$

Since $K \gg 1$, if $R_1/(R_1 + R_2) \gg 1/K$, then

$$e_o = \left(1 + \frac{R_2}{R_1}\right) e_i$$

This equation gives the output voltage e_o. Since e_o and e_i have the same signs, the op-amp circuit shown in Figure 5–45(a) is noninverting.

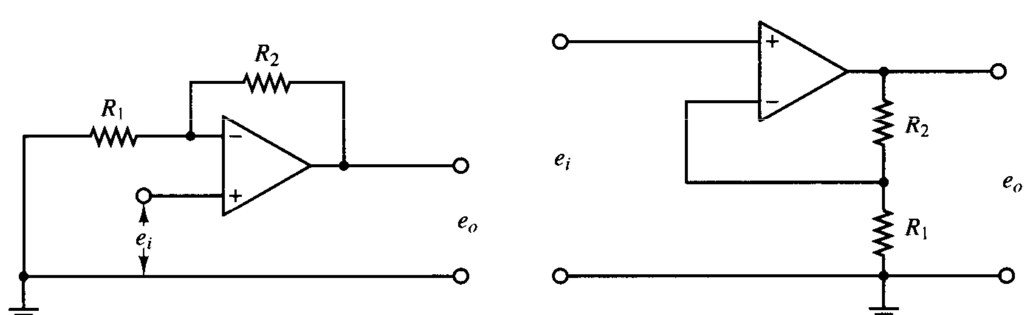


Figure 5–45
(a) Noninverting operational amplifier; (b) equivalent circuit.

EXAMPLE 5-3 Figure 5-46 shows an electrical circuit involving an operational amplifier. Obtain the output e_o.

Let us define

$$i_1 = \frac{e_i - e'}{R_1}, \qquad i_2 = C\frac{d(e' - e_o)}{dt}, \qquad i_3 = \frac{e' - e_o}{R_2}$$

Noting that the current flowing into the amplifier is negligible, we have

$$i_1 = i_2 + i_3$$

Hence

$$\frac{e_i - e'}{R_1} = C\frac{d(e' - e_o)}{dt} + \frac{e' - e_o}{R_2}$$

Since $e' \doteqdot 0$, we have

$$\frac{e_i}{R_1} = -C\frac{de_o}{dt} - \frac{e_o}{R_2}$$

Taking the Laplace transform of this last equation, assuming the zero initial condition, we have

$$\frac{E_i(s)}{R_1} = -\frac{R_2Cs + 1}{R_2}E_o(s)$$

which can be written as

$$\frac{E_o(s)}{E_i(s)} = -\frac{R_2}{R_1}\frac{1}{R_2Cs + 1}$$

The op-amp circuit shown in Figure 5-46 is a first-order lag circuit. (Several other circuits involving op amps are shown in Table 5-1 together with their transfer functions.)

Impedance approach for obtaining transfer functions. Consider the op-amp circuit shown in Figure 5-47. Similar to the case of electrical circuits we discussed earlier, the impedance approach can be applied to op-amp circuits to obtain their transfer functions. For the circuit shown in Figure 5-47, we have

$$E_i(s) = Z_1(s)I(s), \qquad E_o(s) = -Z_2(s)I(s)$$

Hence, the transfer function for the circuit is obtained as

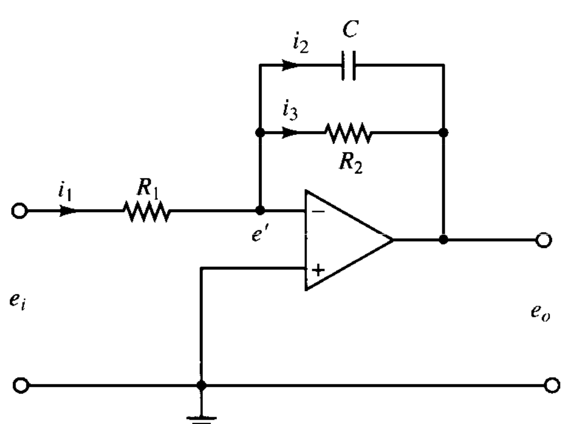


Figure 5-46
First-order lag circuit using operational amplifier.

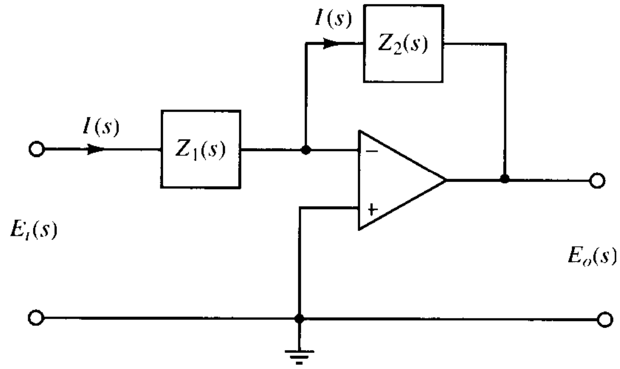


Figure 5-47
Operational amplifier circuit.

$$\frac{E_o(s)}{E_i(s)} = -\frac{Z_2(s)}{Z_1(s)}$$

EXAMPLE 5-4 Referring to the op-amp circuit shown in Figure 5-46, obtain the transfer function $E_o(s)/E_i(s)$ by use of the impedance approach.

The complex impedances $Z_1(s)$ and $Z_2(s)$ for this circuit are

$$Z_1(s) = R_1 \qquad \text{and} \qquad Z_2(s) = \frac{1}{Cs + \frac{1}{R_2}} = \frac{R_2}{R_2Cs + 1}$$

Hence, $E_i(s)$ and $E_o(s)$ are obtained as

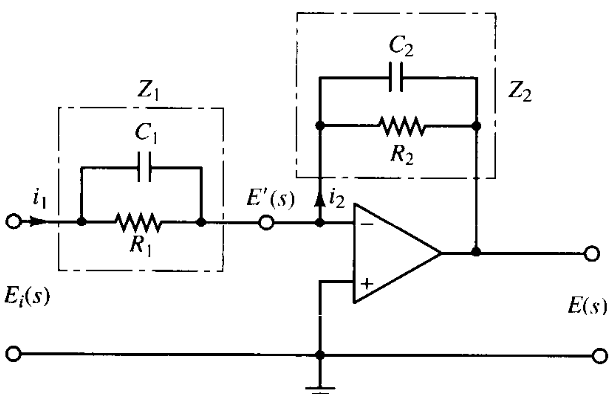
$$E_i(s) = R_1 I(s), \qquad E_o(s) = -\frac{R_2}{R_2Cs + 1} I(s)$$

The transfer function $E_o(s)/E_i(s)$ is, therefore, obtained as

$$\frac{E_o(s)}{E_i(s)} = -\frac{R_2}{R_1} \frac{1}{R_2Cs + 1}$$

which is, of course, the same as that obtained in Example 5-3.

Lead or lag networks using operational amplifiers. Figure 5-48(a) shows an electronic circuit using an operational amplifier. The transfer function for this circuit



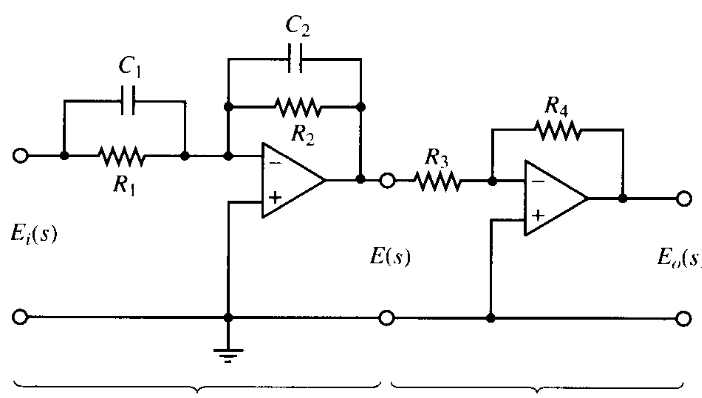


Figure 5-48
(a) Operational-amplifier circuit; (b) operational-amplifier circuit used as a lead or lag compensator.

can be obtained as follows: Define the input impedance and feedback impedance as Z_1 and Z_2, respectively. Then

$$Z_1 = \frac{R_1}{R_1 C_1 s + 1}, \qquad Z_2 = \frac{R_2}{R_2 C_2 s + 1}$$

Since the current flowing into the amplifier is negligible, current i_1 is equal to current i_2. Thus $i_1 = i_2$, or

$$\frac{E_i(s) - E'(s)}{Z_1} = \frac{E'(s) - E(s)}{Z_2}$$

Since $E'(s) \doteqdot 0$, we have

$$\frac{E(s)}{E_i(s)} = -\frac{Z_2}{Z_1} = -\frac{R_2}{R_1}\frac{R_1 C_1 s + 1}{R_2 C_2 s + 1} = -\frac{C_1}{C_2}\frac{s + \dfrac{1}{R_1 C_1}}{s + \dfrac{1}{R_2 C_2}} \tag{5–28}$$

Notice that the transfer function in Equation (5–28) contains a minus sign. Thus, this circuit is sign inverting. If such a sign inversion is not convenient in the actual application, a sign inverter may be connected to either the input or the output of the circuit of Figure 5–48(a). An example is shown in Figure 5–48(b). The sign inverter has the transfer function of

$$\frac{E_o(s)}{E(s)} = -\frac{R_4}{R_3}$$

The sign inverter has the gain of $-R_4/R_3$. Hence the network shown in Figure 5–48(b) has the following transfer function:

$$\frac{E_o(s)}{E_i(s)} = \frac{R_2 R_4}{R_1 R_3}\frac{R_1 C_1 s + 1}{R_2 C_2 s + 1} = \frac{R_4 C_1}{R_3 C_2}\frac{s + \dfrac{1}{R_1 C_1}}{s + \dfrac{1}{R_2 C_2}}$$

$$= K_c \alpha \frac{Ts + 1}{\alpha Ts + 1} = K_c \frac{s + \dfrac{1}{T}}{s + \dfrac{1}{\alpha T}} \tag{5–29}$$

where

$$T = R_1 C_1, \qquad \alpha T = R_2 C_2, \qquad K_c = \frac{R_4 C_1}{R_3 C_2}$$

Notice that

$$K_c \alpha = \frac{R_4 C_1}{R_3 C_2}\frac{R_2 C_2}{R_1 C_1} = \frac{R_2 R_4}{R_1 R_3}, \qquad \alpha = \frac{R_2 C_2}{R_1 C_1}$$

This network has a dc gain of $K_c \alpha = R_2 R_4/(R_1 R_3)$.

Referring to Equation (5–29), this network is a lead network if $R_1C_1 > R_2C_2$, or $a < 1$. It is a lag network if $R_1C_1 < R_2C_2$. (For the definitions of lead and lag networks, refer to Section 5–9.)

PID controller using operational amplifiers. Figure 5–49 shows an electronic proportional-plus-integral-plus-derivative controller (a PID controller) using operational amplifiers. The transfer function $E(s)/E_i(s)$ is given by

$$\frac{E(s)}{E_i(s)} = -\frac{Z_2}{Z_1}$$

where

$$Z_1 = \frac{R_1}{R_1C_1s + 1}, \qquad Z_2 = \frac{R_2C_2s + 1}{C_2s}$$

Thus

$$\frac{E(s)}{E_i(s)} = -\left(\frac{R_2C_2s + 1}{C_2s}\right)\left(\frac{R_1C_1s + 1}{R_1}\right)$$

Noting that

$$\frac{E_o(s)}{E(s)} = -\frac{R_4}{R_3}$$

we have

$$\begin{aligned}
\frac{E_o(s)}{E_i(s)} &= \frac{E_o(s)}{E(s)}\frac{E(s)}{E_i(s)} = \frac{R_4R_2}{R_3R_1}\frac{(R_1C_1s + 1)(R_2C_2s + 1)}{R_2C_2s} \\
&= \frac{R_4R_2}{R_3R_1}\left(\frac{R_1C_1 + R_2C_2}{R_2C_2} + \frac{1}{R_2C_2s} + R_1C_1s\right) \\
&= \frac{R_4(R_1C_1 + R_2C_2)}{R_3R_1C_2}\left[1 + \frac{1}{(R_1C_1 + R_2C_2)s} + \frac{R_1C_1R_2C_2}{R_1C_1 + R_2C_2}s\right] \qquad (5\text{–}30)
\end{aligned}$$

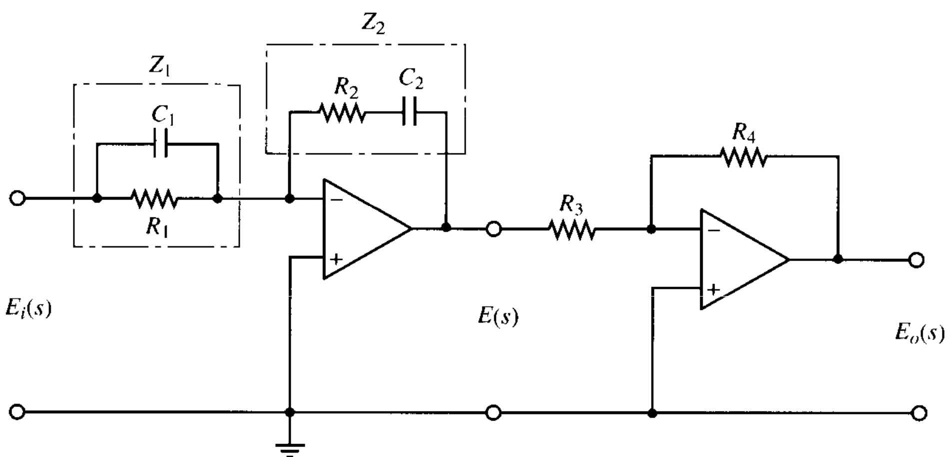


Figure 5–49 Electronic PID controller.

Thus

$$K_p = \frac{R_4(R_1C_1 + R_2C_2)}{R_3R_1C_2}$$

$$T_i = R_1C_1 + R_2C_2$$

$$T_d = \frac{R_1C_1R_2C_2}{R_1C_1 + R_2C_2}$$

In terms of the proportional gain, integral gain, and derivative gain, we have

$$K_p = \frac{R_4(R_1C_1 + R_2C_2)}{R_3R_1C_2}$$

$$K_i = \frac{R_4}{R_3R_1C_2}$$

$$K_d = \frac{R_4R_2C_1}{R_3}$$

Notice that the second operational-amplifier circuit acts as a sign inverter as well as a gain adjuster.

Table 5–1 shows a list of operational-amplifier circuits that may be used as controllers or compensators.

5–9 PHASE LEAD AND PHASE LAG IN SINUSOIDAL RESPONSE

For a sinusoidal input, the steady-state output of a linear time-invariant system is sinusoidal with a phase shift that is a function of the input frequency. This phase angle varies as the frequency is increased from zero to infinity. If the steady-state sinusoidal output of a network leads (lags) the input sinusoid, it is called a lead (lag) network. We shall first derive the steady-state output of a linear, time-invariant network to a sinusoidal input.

Obtaining steady-state outputs to sinusoidal inputs. We shall show that the steady-state output of a transfer function system can be obtained directly from the sinusoidal transfer function, that is, the transfer function in which s is replaced by $j\omega$, where ω is frequency.

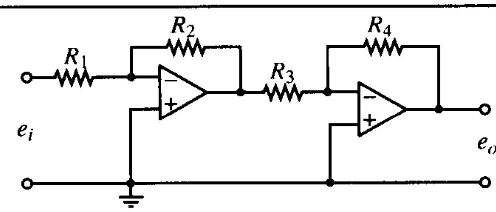
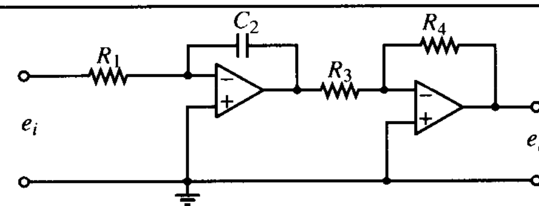
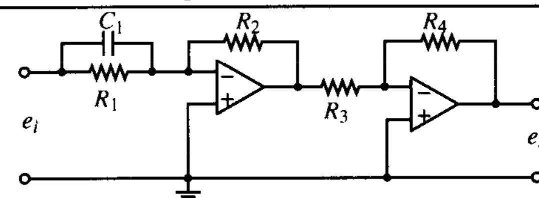
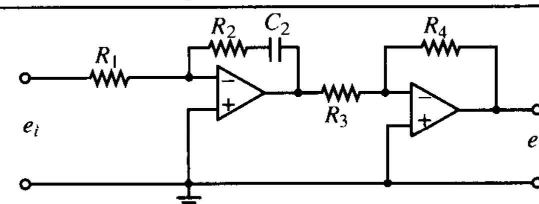
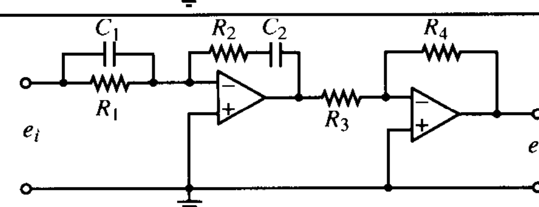
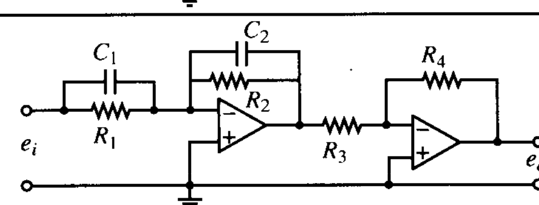
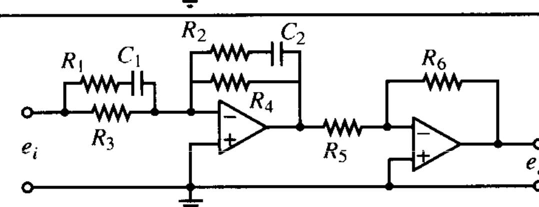
Consider the stable, linear, time-invariant system shown in Figure 5–50. The input and output of the system, whose transfer function is $G(s)$, are denoted by $x(t)$ and $y(t)$, respectively. If the input $x(t)$ is a sinusoidal signal, the steady-state output will also be a sinusoidal signal of the same frequency but with possibly different magnitude and phase angle.

Let us assume that the input signal is given by

$$x(t) = X \sin \omega t$$

Suppose that the transfer function $G(s)$ can be written as a ratio of two polynomials in s; that is,

Table 5–1 Operational-Amplifier Circuits That May Be Used as Compensators

	Control Action	$G(s) = \dfrac{E_o(s)}{E_i(s)}$	Operational Amplifier Circuits
1	P	$\dfrac{R_4}{R_3}\dfrac{R_2}{R_1}$	
2	I	$\dfrac{R_4}{R_3}\dfrac{1}{R_1C_2s}$	
3	PD	$\dfrac{R_4}{R_3}\dfrac{R_2}{R_1}(R_1C_1s+1)$	
4	PI	$\dfrac{R_4}{R_3}\dfrac{R_2}{R_1}\dfrac{R_2C_2s+1}{R_2C_2s}$	
5	PID	$\dfrac{R_4}{R_3}\dfrac{R_2}{R_1}\dfrac{(R_1C_1s+1)(R_2C_2s+1)}{R_2C_2s}$	
6	Lead or lag	$\dfrac{R_4}{R_3}\dfrac{R_2}{R_1}\dfrac{R_1C_1s+1}{R_2C_2s+1}$	
7	Lag–lead	$\dfrac{R_6}{R_5}\dfrac{R_4}{R_3}\dfrac{[(R_1+R_3)C_1s+1](R_2C_2s+1)}{(R_1C_1s+1)[(R_2+R_4)C_2s+1]}$	

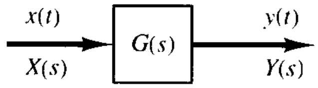


Figure 5–50
Stable, linear, time-invariant system.

$$G(s) = \frac{p(s)}{q(s)} = \frac{p(s)}{(s + s_1)(s + s_2)\cdots(s + s_n)}$$

The Laplace-transformed output $Y(s)$ is then

$$Y(s) = G(s)X(s) = \frac{p(s)}{q(s)}X(s) \tag{5–31}$$

where $X(s)$ is the Laplace transform of the input $x(t)$.

It will be shown that after waiting until steady-state conditions are reached the frequency response can be calculated by replacing s in the transfer function by $j\omega$. It will also be shown that the steady-state response can be given by

$$G(j\omega) = Me^{j\phi} = M\underline{/\phi}$$

where M is the amplitude ratio of the output and input sinusoids and ϕ is the phase shift between the input sinusoid and the output sinusoid. In the frequency-response test, the input frequency ω is varied until the entire frequency range of interest is covered.

The steady-state response of a stable, linear, time-invariant system to a sinusoidal input does not depend on the initial conditions. (Thus, we can assume the zero initial condition.) If $Y(s)$ has only distinct poles, then the partial fraction expansion of Equation (5–31) yields

$$\begin{aligned} Y(s) &= G(s)X(s) = G(s)\frac{\omega X}{s^2 + \omega^2} \\ &= \frac{a}{s + j\omega} + \frac{\bar{a}}{s - j\omega} + \frac{b_1}{s + s_1} + \frac{b_2}{s + s_2} + \cdots + \frac{b_n}{s + s_n} \end{aligned} \tag{5–32}$$

where a and the b_i (where $i = 1, 2, \ldots, n$) are constants and $\bar{a}$ is the complex conjugate of a. The inverse Laplace transform of Equation (5–32) gives

$$y(t) = ae^{-j\omega t} + \bar{a}e^{j\omega t} + b_1e^{-s_1t} + b_2e^{-s_2t} + \cdots + b_ne^{-s_nt} \qquad (t \geq 0) \tag{5–33}$$

For a stable system, $-s_1, -s_2, \ldots, -s_n$ have negative real parts. Therefore, as t approaches infinity, the terms $e^{-s_1t}, e^{-s_2t}, \ldots$, and e^{-s_nt} approach zero. Thus, all the terms on the right-hand side of Equation (5–33), except the first two, drop out at steady state.

If $Y(s)$ involves multiple poles s_j of multiplicity m_j, then $y(t)$ will involve terms such as $t^{h_j}e^{-s_jt}$ ($h_j = 0, 1, 2, \ldots, m_j - 1$). For a stable system, the terms $t^{h_j}e^{-s_jt}$ approach zero as t approaches infinity.

Thus, regardless of whether the system is of the distinct-pole type, the steady-state response becomes

$$y_{ss}(t) = ae^{-j\omega t} + \bar{a}e^{j\omega t} \tag{5–34}$$

where the constant a can be evaluated from Equation (5–32) as follows:

$$a = G(s)\frac{\omega X}{s^2 + \omega^2}(s + j\omega)\bigg|_{s=-j\omega} = -\frac{XG(-j\omega)}{2j}$$

Note that

$$\bar{a} = G(s)\frac{\omega X}{s^2 + \omega^2}(s - j\omega)\bigg|_{s=j\omega} = \frac{XG(j\omega)}{2j}$$

Since $G(j\omega)$ is a complex quantity, it can be written in the following form:

$$G(j\omega) = |G(j\omega)|e^{j\phi}$$

where $|(Gj\omega)|$ represents the magnitude and ϕ represents the angle of $G(j\omega)$; that is,

$$\phi = \underline{/G(j\omega)} = \tan^{-1}\left[\frac{\text{imaginary part of } G(j\omega)}{\text{real part of } G(j\omega)}\right]$$

The angle ϕ may be negative, positive, or zero. Similarly, we obtain the following expression for $G(-j\omega)$:

$$G(-j\omega) = |G(-j\omega)|e^{-j\phi} = |G(j\omega)|e^{-j\phi}$$

Then, noting that

$$a = -\frac{X|G(j\omega)|e^{-j\phi}}{2j}, \qquad \bar{a} = \frac{X|G(j\omega)|e^{j\phi}}{2j}$$

Equation (5–34) can be written

$$\begin{aligned} y_{ss}(t) &= X|G(j\omega)|\frac{e^{j(\omega t+\phi)} - e^{-j(\omega t+\phi)}}{2j} \\ &= X|G(j\omega)|\sin(\omega t + \phi) \\ &= Y\sin(\omega t + \phi) \end{aligned} \tag{5–35}$$

where $Y = X|G(j\omega)|$. We see that a stable, linear, time-invariant system subjected to a sinusoidal input will, at steady state, have a sinusoidal output of the same frequency as the input. But the amplitude and phase of the output will, in general, be different from those of the input. In fact, the amplitude of the output is given by the product of that of the input and $|G(j\omega)|$, while the phase angle differs from that of the input by the amount $\phi = \underline{/G(j\omega)}$. An example of input and output sinusoidal signals is shown in Figure 5–51.

On the basis of this, we obtain this important result: For sinusoidal inputs,

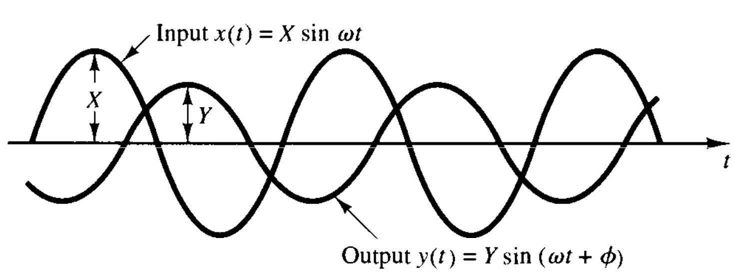


Figure 5–51 Input and output sinusoidal signals.

$$|G(j\omega)| = \left|\frac{Y(j\omega)}{X(j\omega)}\right| = \begin{array}{l}\text{amplitude ratio of the output sinuisoid to the}\\ \text{input sinusoid}\end{array}$$

$$\angle G(j\omega) = \angle \frac{Y(j\omega)}{X(j\omega)} = \begin{array}{l}\text{phase shift of the output sinusoid with respect}\\ \text{to the input sinusoid}\end{array}$$

Hence, the response characteristics of a system to a sinusoidal input can be obtained directly from

$$\frac{Y(j\omega)}{X(j\omega)} = G(j\omega)$$

The function $G(j\omega)$ is called the *sinusoidal transfer function*. It is the ratio of $Y(j\omega)$ to $X(j\omega)$, is a complex quantity, and can be represented by the magnitude and phase angle with frequency as a parameter. (A negative phase angle is called *phase lag*, and a positive phase angle is called *phase lead*.) The sinusoidal transfer function of any linear system is obtained by substituting $j\omega$ for s in the transfer function of the system.

A network that has phase-lead characteristics is commonly called a lead network. Similarly, a network that has phase-lag characteristics is called a lag network.

EXAMPLE 5–5 Consider the system shown in Figure 5–52. The transfer function $G(s)$ is

$$G(s) = \frac{K}{Ts + 1}$$

For the sinusoidal input $x(t) = X \sin \omega t$, the steady-state output $y_{ss}(t)$ can be found as follows: Substituting $j\omega$ for s in $G(s)$ yields

$$G(j\omega) = \frac{K}{jT\omega + 1}$$

The amplitude ratio of the output to input is

$$|G(j\omega)| = \frac{K}{\sqrt{1 + T^2\omega^2}}$$

while the phase angle ϕ is

$$\phi = \angle G(j\omega) = -\tan^{-1} T\omega$$

Thus, for the input $x(t) = X \sin \omega t$, the steady-state output $y_{ss}(t)$ can be obtained from Equation (5–35) as follows:

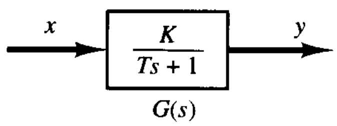


Figure 5–52
First-order system.

$$y_{ss}(t) = \frac{XK}{\sqrt{1 + T^2\omega^2}} \sin(\omega t - \tan^{-1} T\omega) \tag{5-36}$$

From Equation (5–36) it can be seen that for small ω the amplitude of the steady-state output $y_{ss}(t)$ is almost equal to K times the amplitude of the input. The phase shift of the output is small for small ω. For large ω, the amplitude of the output is small and almost inversely proportional to ω. The phase shift approaches $-90°$ as ω approaches infinity. This is a phase-lag network.

EXAMPLE 5–6 Consider the network given by

$$G(s) = \frac{s + \dfrac{1}{T_1}}{s + \dfrac{1}{T_2}}$$

Find if this network is a lead network or lag network.

For the sinusoidal input $x(t) = X \sin \omega t$, the steady-state output $y_{ss}(t)$ can be found as follows: Since

$$G(j\omega) = \frac{j\omega + \dfrac{1}{T_1}}{j\omega + \dfrac{1}{T_2}} = \frac{T_2(1 + T_1 j\omega)}{T_1(1 + T_2 j\omega)}$$

we have

$$|G(j\omega)| = \frac{T_2\sqrt{1 + T_1^2\omega^2}}{T_1\sqrt{1 + T_2^2\omega^2}}$$

and

$$\phi = \underline{/G(j\omega)} = \tan^{-1} T_1\omega - \tan^{-1} T_2\omega$$

Thus the steady-state output is

$$y_{ss}(t) = \frac{XT_2\sqrt{1 + T_1^2\omega^2}}{T_1\sqrt{1 + T_2^2\omega^2}} \sin(\omega t + \tan^{-1} T_1\omega - \tan^{-1} T_2\omega)$$

From this expression, we find that if $T_1 > T_2$ then $\tan^{-1} T_1\omega - \tan^{-1} T_2\omega > 0$. Thus, if $T_1 > T_2$, then the network is a lead network. If $T_1 < T_2$, then the network is a lag network.

5–10 STEADY-STATE ERRORS IN UNITY-FEEDBACK CONTROL SYSTEMS

Errors in a control system can be attributed to many factors. Changes in the reference input will cause unavoidable errors during transient periods and may also cause steady-state errors. Imperfections in the system components, such as static friction, backlash, and amplifier drift, as well as aging or deterioration, will cause errors at steady state. In this section, however, we shall not discuss errors due to imperfections in the system components. Rather, we shall investigate a type of steady-state error that is caused by the incapability of a system to follow particular types of inputs.

Any physical control system inherently suffers steady-state error in response to certain types of inputs. A system may have no steady-state error to a step input, but the same system may exhibit nonzero steady-state error to a ramp input. (The only way we may be able to eliminate this error is to modify the system structure.) Whether a given system will exhibit steady-state error for a given type of input depends on the type of open-loop transfer function of the system, to be discussed in what follows.

Classification of control systems. Control systems may be classified according to their ability to follow step inputs, ramp inputs, parabolic inputs, and so on. This is a reasonable classification scheme because actual inputs may frequently be considered combinations of such inputs. The magnitudes of the steady-state errors due to these individual inputs are indicative of the goodness of the system.

Consider the unity-feedback control system with the following open-loop transfer function $G(s)$:

$$G(s) = \frac{K(T_a s + 1)(T_b s + 1)\cdots(T_m s + 1)}{s^N(T_1 s + 1)(T_2 s + 1)\cdots(T_p s + 1)}$$

It involves the term s^N in the denominator, representing a pole of multiplicity N at the origin. The present classification scheme is based on the number of integrations indicated by the open-loop transfer function. A system is called type 0, type 1, type 2, ..., if $N = 0, N = 1, N = 2, \ldots$, respectively. Note that this classification is different from that of the order of a system. As the type number is increased, accuracy is improved; however, increasing the type number aggravates the stability problem. A compromise between steady-state accuracy and relative stability is always necessary. In practice, it is rather exceptional to have type 3 or higher systems because we find it generally difficult to design stable systems having more than two integrations in the feedforward path.

We shall see later that, if $G(s)$ is written so that each term in the numerator and denominator, except the term s^N, approaches unity as s approaches zero, then the open-loop gain K is directly related to the steady-state error.

Steady-state errors. Consider the system shown in Figure 5–53. The closed-loop transfer function is

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)}$$

The transfer function between the error signal $e(t)$ and the input signal $r(t)$ is

$$\frac{E(s)}{R(s)} = 1 - \frac{C(s)}{R(s)} = \frac{1}{1 + G(s)}$$

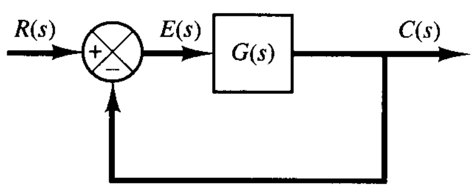


Figure 5–53
Control system.

where the error $e(t)$ is the difference between the input signal and the output signal.

The final-value theorem provides a convenient way to find the steady-state performance of a stable system. Since $E(s)$ is

$$E(s) = \frac{1}{1 + G(s)} R(s)$$

the steady-state error is

$$e_{ss} = \lim_{t \to \infty} e(t) = \lim_{s \to 0} sE(s) = \lim_{s \to 0} \frac{sR(s)}{1 + G(s)}$$

The static error constants defined in the following are figures of merit of control systems. The higher the constants, the smaller the steady-state error. In a given system, the output may be the position, velocity, pressure, temperature, or the like. The physical form of the output, however, is immaterial to the present analysis. Therefore, in what follows, we shall call the output "position," the rate of change of the output "velocity," and so on. This means that in a temperature control system "position" represents the output temperature, "velocity" represents the rate of change of the output temperature, and so on.

Static position error constant K_p. The steady-state error of the system for a unit-step input is

$$\begin{aligned} e_{ss} &= \lim_{s \to 0} \frac{s}{1 + G(s)} \frac{1}{s} \\ &= \frac{1}{1 + G(0)} \end{aligned}$$

The static position error constant K_p is defined by

$$K_p = \lim_{s \to 0} G(s) = G(0)$$

Thus, the steady-state error in terms of the static position error constant K_p is given by

$$e_{ss} = \frac{1}{1 + K_p}$$

For a type 0 system,

$$K_p = \lim_{s \to 0} \frac{K(T_a s + 1)(T_b s + 1) \cdots}{(T_1 s + 1)(T_2 s + 1) \cdots} = K$$

For a type 1 or higher system,

$$K_p = \lim_{s \to 0} \frac{K(T_a s + 1)(T_b s + 1) \cdots}{s^N (T_1 s + 1)(T_2 s + 1) \cdots} = \infty, \qquad \text{for } N \geq 1$$

Hence, for a type 0 system, the static position error constant K_p is finite, while for a type 1 or higher system, K_p is infinite.

For a unit-step input, the steady-state error e_{ss} may be summarized as follows:

$$e_{ss} = \frac{1}{1 + K}, \qquad \text{for type 0 systems}$$

$$e_{ss} = 0, \qquad \text{for type 1 or higher systems}$$

From the foregoing analysis, it is seen that the response of a feedback control system to a step input involves a steady-state error if there is no integration in the feedforward path. (If small errors for step inputs can be tolerated, then a type 0 system may be permissible, provided that the gain K is sufficiently large. If the gain K is too large, however, it is difficult to obtain reasonable relative stability.) If zero steady-state error for a step input is desired, the type of the system must be one or higher.

Static velocity error constant K_v. The steady-state error of the system with a unit-ramp input is given by

$$e_{ss} = \lim_{s \to 0} \frac{s}{1 + G(s)} \frac{1}{s^2}$$

$$= \lim_{s \to 0} \frac{1}{sG(s)}$$

The static velocity error constant K_v is defined by

$$K_v = \lim_{s \to 0} sG(s)$$

Thus, the steady-state error in terms of the static velocity error constant K_v is given by

$$e_{ss} = \frac{1}{K_v}$$

The term *velocity error* is used here to express the steady-state error for a ramp input. The dimension of the velocity error is the same as the system error. That is, velocity error is not an error in velocity, but it is an error in position due to a ramp input.

For a type 0 system,

$$K_v = \lim_{s \to 0} \frac{sK(T_a s + 1)(T_b s + 1) \cdots}{(T_1 s + 1)(T_2 s + 1) \cdots} = 0$$

For a type 1 system,

$$K_v = \lim_{s \to 0} \frac{sK(T_a s + 1)(T_b s + 1) \cdots}{s(T_1 s + 1)(T_2 s + 1) \cdots} = K$$

For a type 2 or higher system,

$$K_v = \lim_{s \to 0} \frac{sK(T_a s + 1)(T_b s + 1) \cdots}{s^N(T_1 s + 1)(T_2 s + 1) \cdots} = \infty, \qquad \text{for } N \geq 2$$

The steady-state error e_{ss} for the unit-ramp input can be summarized as follows:

$$e_{ss} = \frac{1}{K_v} = \infty, \qquad \text{for type 0 systems}$$

$$e_{ss} = \frac{1}{K_v} = \frac{1}{K}, \qquad \text{for type 1 systems}$$

$$e_{ss} = \frac{1}{K_v} = 0, \qquad \text{for type 2 or higher systems}$$

The foregoing analysis indicates that a type 0 system is incapable of following a ramp input in the steady state. The type 1 system with unity feedback can follow the ramp input with a finite error. In steady-state operation, the output velocity is exactly the same as the input velocity, but there is a positional error. This error is proportional to the velocity of the input and is inversely proportional to the gain K. Figure 5–54 shows an example of the response of a type 1 system with unity feedback to a ramp input. The type 2 or higher system can follow a ramp input with zero error at steady state.

Static acceleration error constant K_a. The steady-state error of the system with a unit-parabolic input (acceleration input), which is defined by

$$\begin{aligned} r(t) &= \frac{t^2}{2}, && \text{for } t \geq 0 \\ &= 0, && \text{for } t < 0 \end{aligned}$$

is given by

$$\begin{aligned} e_{ss} &= \lim_{s \to 0} \frac{s}{1 + G(s)} \frac{1}{s^3} \\ &= \frac{1}{\lim_{s \to 0} s^2 G(s)} \end{aligned}$$

The static acceleration error constant K_a is defined by the equation

$$K_a = \lim_{s \to 0} s^2 G(s)$$

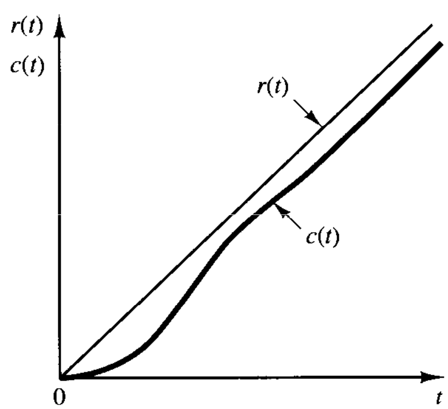


Figure 5–54
Response of a type 1 unity-feedback system to a ramp input.

The steady-state error is then

$$e_{ss} = \frac{1}{K_a}$$

Note that the acceleration error, the steady-state error due to a parabolic input, is an error in position.

The values of K_a are obtained as follows:

For a type 0 system,

$$K_a = \lim_{s\to 0} \frac{s^2 K(T_a s + 1)(T_b s + 1)\cdots}{(T_1 s + 1)(T_2 s + 1)\cdots} = 0$$

For a type 1 system,

$$K_a = \lim_{s\to 0} \frac{s^2 K(T_a s + 1)(T_b s + 1)\cdots}{s(T_1 s + 1)(T_2 s + 1)\cdots} = 0$$

For a type 2 system,

$$K_a = \lim_{s\to 0} \frac{s^2 K(T_a s + 1)(T_b s + 1)\cdots}{s^2(T_1 s + 1)(T_2 s + 1)\cdots} = K$$

For a type 3 or higher system,

$$K_a = \lim_{s\to 0} \frac{s^2 K(T_a s + 1)(T_b s + 1)\cdots}{s^N(T_1 s + 1)(T_2 s + 1)\cdots} = \infty, \qquad \text{for } N \geq 3$$

Thus, the steady-state error for the unit parabolic input is

$$e_{ss} = \infty, \qquad \text{for type 0 and type 1 systems}$$

$$e_{ss} = \frac{1}{K}, \qquad \text{for type 2 systems}$$

$$e_{ss} = 0, \qquad \text{for type 3 or higher systems}$$

Note that both type 0 and type 1 systems are incapable of following a parabolic input in the steady state. The type 2 system with unity feedback can follow a parabolic input with a finite error signal. Figure 5–55 shows an example of the response of a type 2 system with unity feedback to a parabolic input. The type 3 or higher system with unity feedback follows a parabolic input with zero error at steady state.

Summary. Table 5–2 summarizes the steady-state errors for type 0, type 1, and type 2 systems when they are subjected to various inputs. The finite values for steady-state errors appear on the diagonal line. Above the diagonal, the steady-state errors are infinity; below the diagonal, they are zero.

Remember that the terms *position error*, *velocity error*, and *acceleration error* mean steady-state deviations in the output position. A finite velocity error implies that after transients have died out the input and output move at the same velocity but have a finite position difference.

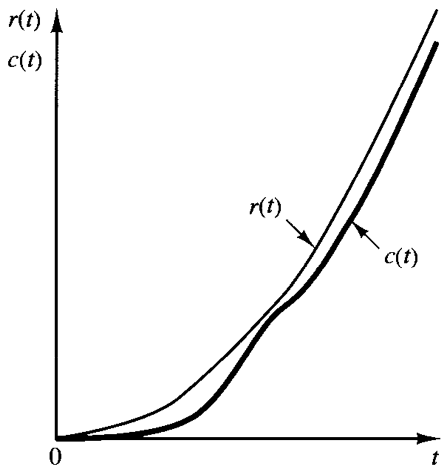


Figure 5–55
Response of a type 2 unity-feedback system to a parabolic input.

The error constants K_p, K_v, and K_a describe the ability of a unity-feedback system to reduce or eliminate steady-state error. Therefore, they are indicative of the steady-state performance. It is generally desirable to increase the error constants, while maintaining the transient response within an acceptable range. If there is any conflict between the static velocity error constant and the static acceleration error constant, then the latter may be considered less important than the former. It is noted that to improve the steady-state performance we can increase the type of the system by adding an integrator or integrators to the feedforward path. This, however, introduces an additional stability problem. The design of a satisfactory system with more than two integrators in series in the feedforward path is generally difficult.

Comparison of steady-state errors in open-loop control system and closed-loop control system. Consider the open-loop control system and closed-loop control system shown in Figure 5–56. In the open loop one, gain K_c is calibrated so that $K_c = 1/K$. Thus, the transfer function of the open-loop control system is

$$G_0(s) = \frac{1}{K}\frac{K}{Ts + 1} = \frac{1}{Ts + 1}$$

In the closed-loop control system, gain K_p of the controller is set so that $K_pK \gg 1$.

Table 5–2 Steady-State Error in Terms of Gain K

	Step Input $r(t) = 1$	Ramp Input $r(t) = t$	Acceleration Input $r(t) = \frac{1}{2}t^2$
Type 0 system	$\frac{1}{1 + K}$	∞	∞
Type 1 system	0	$\frac{1}{K}$	∞
Type 2 system	0	0	$\frac{1}{K}$

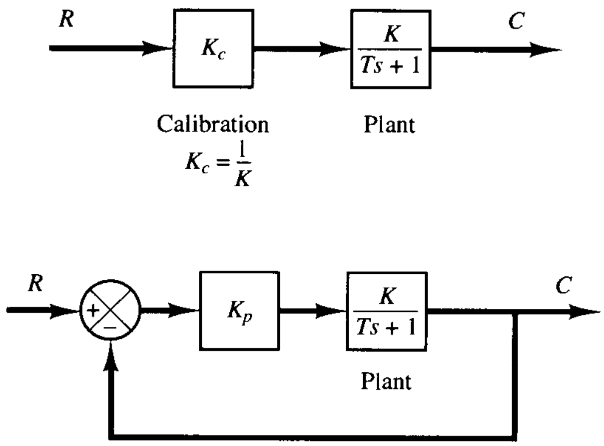


Figure 5–56
Block diagrams of an open-loop control system and a closed-loop control system.

Assuming a unit-step input, let us compare the steady-state errors for these control systems. For the open-loop control system, the error signal is

$$e(t) = r(t) - c(t)$$

or

$$\begin{aligned} E(s) &= R(s) - C(s) \\ &= [1 - G_0(s)]R(s) \end{aligned}$$

The steady-state error in the unit-step response is

$$\begin{aligned} e_{ss} &= \lim_{s \to 0} sE(s) \\ &= \lim_{s \to 0} s[1 - G_0(s)]\frac{1}{s} \\ &= 1 - G_0(0) \end{aligned}$$

If $G_o(0)$, the dc gain of the open-loop control system, is equal to unity, then the steady-state error is zero. Due to environmental changes and aging of components, however, the dc gain $G_o(0)$ will drift from unity as time elapses, and the steady-state error will no longer be equal to zero. Such steady-state error in an open-loop control system will remain until the system is recalibrated.

For the closed-loop control system, the error signal is

$$\begin{aligned} E(s) &= R(s) - C(s) \\ &= \frac{1}{1 + G(s)}R(s) \end{aligned}$$

where

$$G(s) = \frac{K_p K}{Ts + 1}$$

The steady-state error in the unit-step response is

$$e_{ss} = \lim_{s \to 0} s \left[\frac{1}{1 + G(s)} \right] \frac{1}{s}$$
$$= \frac{1}{1 + G(0)}$$
$$= \frac{1}{1 + K_p K}$$

In the closed-loop control system, gain K_p is set at a large value compared with $1/K$. Thus the steady-state error can be made small, although not exactly zero.

Let us assume the following variation in the transfer function of the plant, assuming K_c and K_p constant:

$$\frac{K + \Delta K}{Ts + 1}$$

For simplicity, let us assume that $K = 10, \Delta K = 1$, or $\Delta K/K = 0.1$. Then the steady-state error in the unit-step response for the open-loop control system becomes

$$e_{ss} = 1 - \frac{1}{K}(K + \Delta K)$$
$$= 1 - 1.1 = -0.1$$

For the closed-loop control system, if K_p is set at $100/K$, then the steady-state error in the unit-step response becomes

$$e_{ss} = \frac{1}{1 + G(0)}$$
$$= \frac{1}{1 + \frac{100}{K}(K + \Delta K)}$$
$$= \frac{1}{1 + 110} = 0.009$$

Thus, the closed-loop control system is superior to the open-loop control system in the presence of environmental changes, aging of components, and the like, which definitely affect the steady-state performance.

EXAMPLE PROBLEMS AND SOLUTIONS

A-5-1. Explain why the proportional control of a plant that does not possess an integrating property (which means that the plant transfer function does not include the factor $1/s$) suffers offset in response to step inputs.

Solution. Consider, for example, the system shown in Figure 5–57. At steady state, if c were equal to a nonzero constant r, then $e = 0$ and $u = Ke = 0$, resulting in $c = 0$, which contradicts the assumption that $c = r =$ nonzero constant.

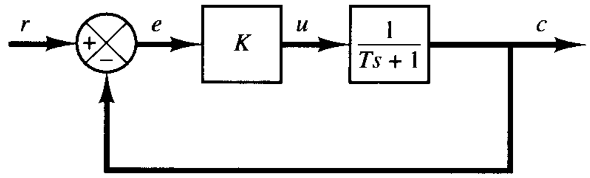


Figure 5–57
Control system.

A nonzero offset must exist for proper operation of such a control system. In other words, at steady state, if e were equal to $r/(1 + K)$, then $u = Kr/(1 + K)$ and $c = Kr/(1 + K)$, which results in the assumed error signal $e = r/(1 + K)$. Thus the offset of $r/(1 + K)$ must exist in such a system.

A-5-2. Consider the system shown in Figure 5–58. Show that the steady-state error in following the unit-ramp input is B/K. This error can be made smaller by choosing B small and/or K large. However, making B small and/or K large would have the effect of making the damping ratio small, which is normally not desirable. Describe a method or methods to make B/K small and yet make the damping ratio have reasonable value $(0.5 < \zeta < 0.7)$.

Solution. From Figure 5–58 we obtain

$$E(s) = R(s) - C(s) = \frac{Js^2 + Bs}{Js^2 + Bs + K} R(s)$$

The steady-state error for the unit-ramp response can be obtained as follows: For the unit-ramp input, the steady-state error e_{ss} is

$$\begin{aligned} e_{ss} &= \lim_{s \to 0} sE(s) \\ &= \lim_{s \to 0} s \frac{Js^2 + Bs}{Js^2 + Bs + K} \frac{1}{s^2} \\ &= \frac{B}{K} \\ &= \frac{2\zeta}{\omega_n} \end{aligned}$$

where

$$\zeta = \frac{B}{2\sqrt{KJ}}, \qquad \omega_n = \sqrt{\frac{K}{J}}$$

To assure acceptable transient response and acceptable steady-state error in following a ramp input, ζ must not be too small and ω_n must be sufficiently large. It is possible to make the steady-state error e_{ss} small by making the value of the gain K large. (A large value of K has an additional advantage of suppressing undesirable effects caused by dead zone, backlash, coulomb friction,

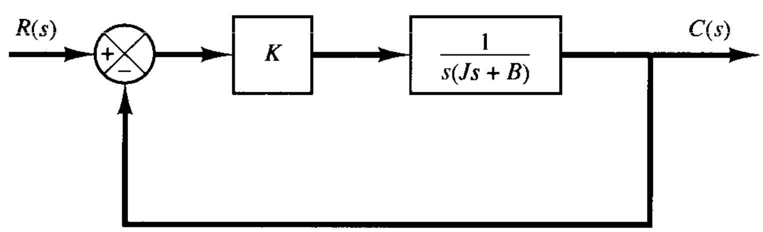


Figure 5–58
Control system.

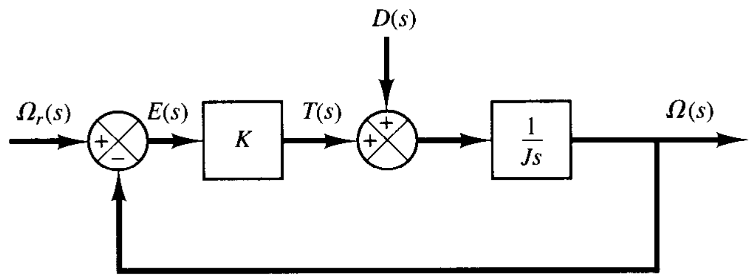


Figure 5–59 Block diagram of a speed control system.

and the like.) A large value of K would, however, make the value of ζ small and increase the maximum overshoot, which is undesirable.

It is therefore necessary to compromise between the magnitude of the steady-state error to a ramp input and the maximum overshoot to a unit-step input. In the system shown in Figure 5–58, a reasonable compromise may not be reached easily. It is then desirable to consider other types of control action that may improve both the transient-response and steady-state behavior. Two schemes to improve both the transient-response and steady-state behavior are available. One scheme is to use a proportional-plus-derivative controller and the other is to use tachometer feedback.

A-5-3. The block diagram of Figure 5–59 shows a speed control system in which the output member of the system is subject to a torque disturbance. In the diagram, $\Omega_r(s)$, $\Omega(s)$, $T(s)$, and $D(s)$ are the Laplace transforms of the reference speed, output speed, driving torque, and disturbance torque, respectively. In the absence of a disturbance torque, the output speed is equal to the reference speed.

Investigate the response of this system to a unit-step disturbance torque. Assume that the reference input is zero, or $\Omega_r(s) = 0$.

Solution. Figure 5–60 is a modified block diagram convenient for the present analysis. The closed-loop transfer function is

$$\frac{\Omega_D(s)}{D(s)} = \frac{1}{Js + K}$$

where $\Omega_D(s)$ is the Laplace transform of the output speed due to the disturbance torque. For a unit-step disturbance torque, the steady-state output velocity is

$$\begin{aligned} \omega_D(\infty) &= \lim_{s \to 0} s\Omega_D(s) \\ &= \lim_{s \to 0} \frac{s}{Js + K} \frac{1}{s} \\ &= \frac{1}{K} \end{aligned}$$

From this analysis, we conclude that, if a step disturbance torque is applied to the output member of the system, an error speed will result so that the ensuing motor torque will exactly

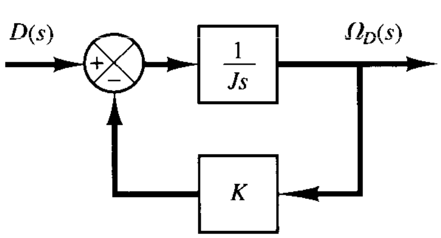


Figure 5–60 Block diagram of the speed control system of Figure 5–59 when $\Omega_r(s) = 0$.

cancel the disturbance torque. To develop this motor torque, it is necessary that there be an error in speed so that nonzero torque will result.

A-5-4. In the system considered in Problem A-5-3, it is desired to eliminate as much as possible the speed errors due to torque disturbances.

Is it possible to cancel the effect of a disturbance torque at steady state so that a constant disturbance torque applied to the output member will cause no speed change at steady state?

Solution. Suppose that we choose a suitable controller whose transfer function is $G_c(s)$, as shown in Figure 5–61. Then in the absence of the reference input the closed-loop transfer function between the output velocity $\Omega_D(s)$ and the disturbance torque $D(s)$ is

$$\frac{\Omega_D(s)}{D(s)} = \frac{\frac{1}{Js}}{1 + \frac{1}{Js} G_c(s)}$$

$$= \frac{1}{Js + G_c(s)}$$

The steady-state output speed due to a unit-step disturbance torque is

$$\omega_D(\infty) = \lim_{s \to 0} s\Omega_D(s)$$

$$= \lim_{s \to 0} \frac{s}{Js + G_c(s)} \frac{1}{s}$$

$$= \frac{1}{G_c(0)}$$

To satisfy the requirement that

$$\omega_D(\infty) = 0$$

we must choose $G_c(0) = \infty$. This can be realized if we choose

$$G_c(s) = \frac{K}{s}$$

Integral control action will continue to correct until the error is zero. This controller, however, presents a stability problem because the characteristic equation will have two imaginary roots.

One method of stabilizing such a system is to add a proportional mode to the controller or choose

$$G_c(s) = K_p + \frac{K}{s}$$

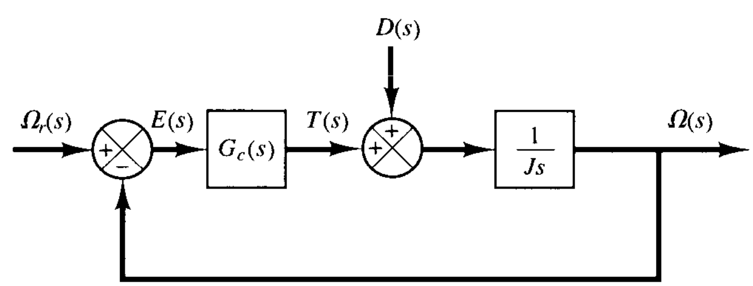


Figure 5–61
Block diagram of a speed control system.

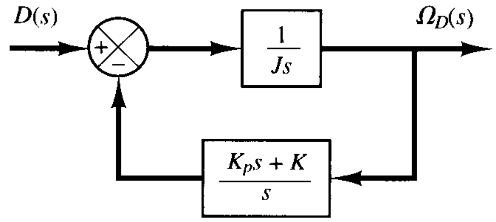


Figure 5–62
Block diagram of the speed control system of Figure 5–61 when $G_c(s) = K_p + (K/s)$ and $\Omega_r(s) = 0$.

With this controller, the block diagram of Figure 5–61 in the absence of the reference input can be modified to that of Figure 5–62. The closed-loop transfer function $\Omega_D(s)/D(s)$ becomes

$$\frac{\Omega_D(s)}{D(s)} = \frac{s}{Js^2 + K_p s + K}$$

For a unit-step disturbance torque, the steady-state output speed is

$$\omega_D(\infty) = \lim_{s \to 0} s\Omega_D(s) = \lim_{s \to 0} \frac{s^2}{Js^2 + K_p s + K} \frac{1}{s} = 0$$

Thus, we see that the proportional-plus-integral controller eliminates speed error at steady state.

The use of integral control action has increased the order of the system by 1. (This tends to produce an oscillatory response.)

In the present system, a step disturbance torque will cause a transient error in the output speed, but the error will become zero at steady state. The integrator provides a nonzero output with zero error. (The nonzero output of the integrator produces a motor torque that exactly cancels the disturbance torque.)

Note that the integrator in the transfer function of the plant does not eliminate the steady-state error due to a step disturbance torque. To eliminate this, we must have an integrator before the point where the disturbance torque enters.

A-5-5. Consider the system shown in Figure 5–63(a). The steady-state error to a unit-ramp input is $e_{ss} = 2\zeta/\omega_n$. Show that the steady-state error for following a ramp input may be eliminated if the input is introduced to the system through a proportional-plus-derivative filter, as shown in Figure 5–63(b), and the value of k is properly set. Note that the error $e(t)$ is given by $r(t) - c(t)$.

Solution. The closed-loop transfer function of the system shown in Figure 5–63(b) is

$$\frac{C(s)}{R(s)} = \frac{(1 + ks)\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Then

$$R(s) - C(s) = \left(\frac{s^2 + 2\zeta\omega_n s - \omega_n^2 ks}{s^2 + 2\zeta\omega_n s + \omega_n^2}\right) R(s)$$

If the input is a unit ramp, then the steady-state error is

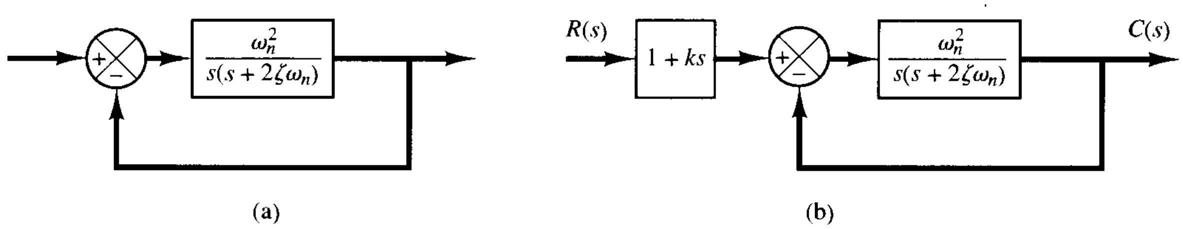


Figure 5–63
(a) Control system;
(b) control system with input filter.

$$e(\infty) = r(\infty) - c(\infty)$$
$$= \lim_{s\to 0} s\left(\frac{s^2 + 2\zeta\omega_n s - \omega_n^2 ks}{s^2 + 2\zeta\omega_n s + \omega_n^2}\right)\frac{1}{s^2}$$
$$= \frac{2\zeta\omega_n - \omega_n^2 k}{\omega_n^2}$$

Therefore, if k is chosen as

$$k = \frac{2\zeta}{\omega_n}$$

then the steady-state error for following a ramp input can be made equal to zero. Note that, if there are any variations in the values of ζ and/or ω_n due to environmental changes or aging, then a nonzero steady-state error for a ramp response may result.

A-5-6. Consider the liquid-level control system shown in Figure 5–64. Assume that the set point of the controller is fixed. Assuming a step disturbance of magnitude D_0, determine the error. Assume that D_0 is small and the variations in the variables from their respective steady-state values are also small. The controller is proportional.

If the controller is not proportional, but integral, what is the steady-state error?

Solution. Figure 5–65 is a block diagram of the system when the controller is proportional with gain K_p. (We assume the transfer function of the pneumatic valve to be unity.) Since the set point is fixed, the variation in the set point is zero, or $X(s) = 0$. The Laplace transform of $h(t)$ is

$$H(s) = \frac{K_p R}{RCs + 1}E(s) + \frac{R}{RCs + 1}D(s)$$

Then

$$E(s) = -H(s) = -\frac{K_p R}{RCs + 1}E(s) - \frac{R}{RCs + 1}D(s)$$

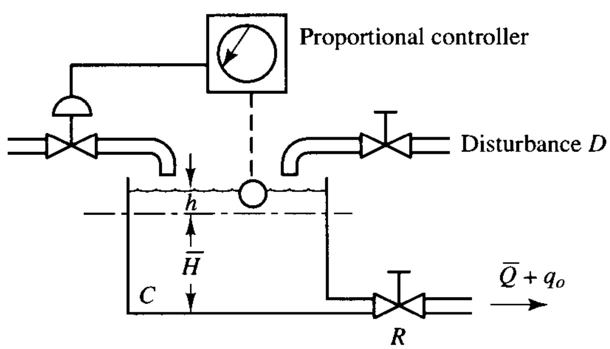


Figure 5–64
Liquid-level control system.

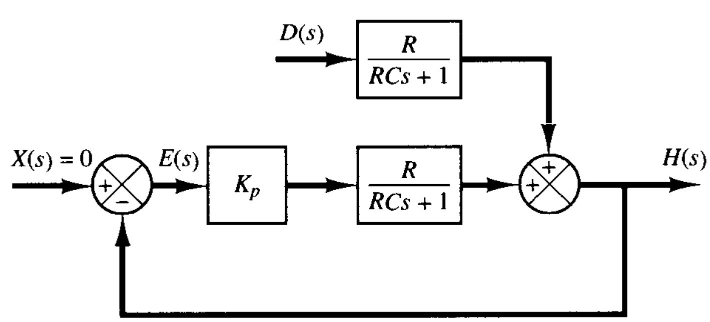


Figure 5–65
Block diagram of the liquid-level control system shown in Figure 5–64.

Hence

$$E(s) = -\frac{R}{RCs + 1 + K_pR} D(s)$$

Since

$$D(s) = \frac{D_0}{s}$$

we obtain

$$\begin{aligned} E(s) &= -\frac{R}{RCs + 1 + K_pR} \frac{D_0}{s} \\ &= \frac{RD_0}{1 + K_pR} \left(\frac{1}{s + \dfrac{1 + K_pR}{RC}} \right) - \frac{RD_0}{1 + K_pR} \frac{1}{s} \end{aligned}$$

The time solution for $t > 0$ is

$$e(t) = \frac{RD_0}{1 + K_pR} \left[\exp\left(-\frac{1 + K_pR}{RC} t \right) - 1 \right]$$

Thus, the time constant is $RC/(1 + K_pR)$. (In the absence of the controller, the time constant is equal to RC.) As the gain of the controller is increased, the time constant is decreased. The steady-state error is

$$e(\infty) = -\frac{RD_0}{1 + K_pR}$$

As the gain K_p of the controller is increased, the steady-state error, or offset, is reduced. Thus, mathematically, the larger the gain K_p is, the smaller the offset and time constant are. In practical systems, however, if the gain K_p of the proportional controller is increased to a very large value, oscillation may result in the output since in our analysis all the small lags and small time constants that may exist in the actual control system are neglected. (If these small lags and time constants are included in the analysis, the transfer function becomes higher order, and for very large values of K_p the possibility of oscillation or even instability may occur.)

If the controller is integral, then assuming the transfer function of the controller to be

$$G_c = \frac{K}{s}$$

we obtain

$$E(s) = -\frac{Rs}{RCs^2 + s + KR} D(s)$$

The steady-state error for a step disturbance $D(s) = D_0/(s)$ is

$$\begin{aligned} e(\infty) &= \lim_{s \to 0} sE(s) \\ &= \lim_{s \to 0} \frac{-Rs^2}{RCs^2 + s + KR} \frac{D_0}{s} \\ &= 0 \end{aligned}$$

Thus, an integral controller eliminates steady-state error or offset due to the step disturbance. (The value of K must be chosen so that the transient response due to the command input and/or disturbance damps out with a reasonable speed.)

A-5-7. Obtain both analytical and computational solutions of the unit-step response of a unity-feedback system whose open-loop transfer function is

$$G(s) = \frac{5(s + 20)}{s(s + 4.59)(s^2 + 3.41s + 16.35)}$$

Solution. The closed-loop transfer function is

$$\frac{C(s)}{R(s)} = \frac{5(s + 20)}{s(s + 4.59)(s^2 + 3.41s + 16.35) + 5(s + 20)}$$

$$= \frac{5s + 100}{s^4 + 8s^3 + 32s^2 + 80s + 100}$$

$$= \frac{5(s + 20)}{(s^2 + 2s + 10)(s^2 + 6s + 10)}$$

The unit-step response of this system is then

$$C(s) = \frac{5(s + 20)}{s(s^2 + 2s + 10)(s^2 + 6s + 10)}$$

$$= \frac{1}{s} + \frac{\frac{3}{8}(s + 1) - \frac{17}{8}}{(s + 1)^2 + 3^2} + \frac{-\frac{11}{8}(s + 3) - \frac{13}{8}}{(s + 3)^2 + 1^2}$$

The time response $c(t)$ can be found by taking the inverse Laplace transform of $C(s)$ as follows:

$$c(t) = 1 + \tfrac{3}{8}e^{-t}\cos 3t - \tfrac{17}{24}e^{-t}\sin 3t - \tfrac{11}{8}e^{-3t}\cos t - \tfrac{13}{8}e^{-3t}\sin t, \qquad \text{for } t \geq 0$$

A MATLAB program to obtain the unit-step response of this system is shown in MATLAB Program 5–2. The resulting unit-step response curve is shown in Figure 5–66.

MATLAB Program 5–2

```
% ---------- Unit-step-response ----------

num = [0   0   0   5   100];
den = [1   8   32   80   100];
step(num,den)
grid
title('Unit-Step Response of C(s)/R(s) = (5s + 100)/(s^4 + 8s^3 + 32s^2 + 80s + 100)')
```

A-5-8. Consider the following characteristic equation:

$$s^4 + Ks^3 + s^2 + s + 1 = 0$$

Determine the range of K for stability.

Solution. The Routh array of coefficients is

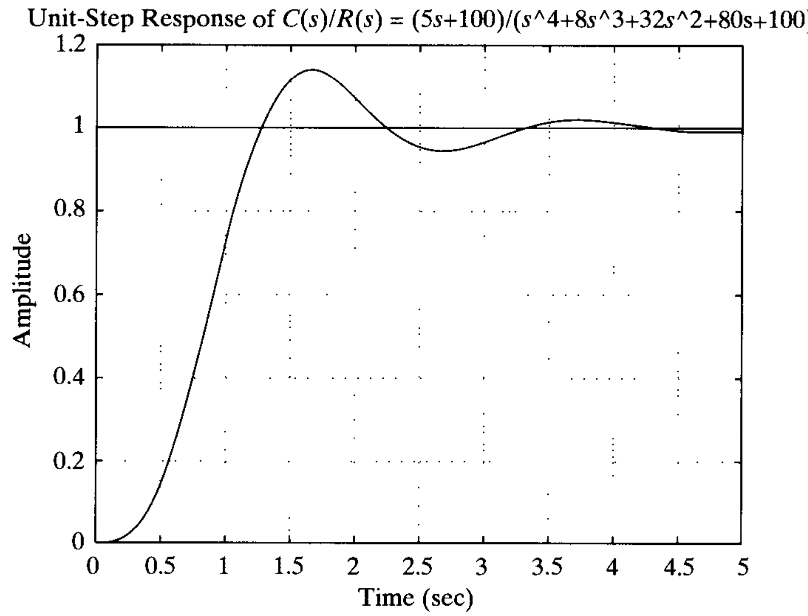


Figure 5–66
Unit-step response curve.

$$
\begin{array}{llll}
s^4 & 1 & 1 & 1 \\
s^3 & K & 1 & 0 \\
s^2 & \dfrac{K-1}{K} & 1 & 0 \\
s^1 & 1-\dfrac{K^2}{K-1} & 0 & \\
s^0 & 1 & &
\end{array}
$$

For stability, we require that

$$K > 0$$

$$\frac{K-1}{K} > 0$$

$$1 - \frac{K^2}{K-1} > 0$$

From the first and second conditions, K must be greater than 1. For $K > 1$, notice that the term $1 - [K^2/(K-1)]$ is always negative, since

$$\frac{K - 1 - K^2}{K-1} = \frac{-1 + K(1-K)}{K-1} < 0$$

Thus, the three conditions cannot be fulfilled simultaneously. Therefore, there is no value of K that allows stability of the system.

A-5-9. Consider the characteristic equation given by

$$a_0 s^n + a_1 s^{n-1} + a_2 s^{n-2} + \cdots + a_{n-1}s + a_n = 0 \tag{5–37}$$

The Hurwitz stability criterion, given next, gives conditions for all the roots to have negative real parts in terms of the coefficients of the polynomial. As stated in the discussions of Routh's stability criterion in Section 5–5, for all the roots to have negative real parts, all the coefficients a's must be positive. This is a necessary condition but not a sufficient condition. If this condition is not satisfied, it indicates that some of the roots have positive real parts or are imaginary or zero. A sufficient condition for all the roots to have negative real parts is given in the following Hurwitz stability criterion: If all the coefficients of the polynomial are positive, arrange these coefficients in the following determinant:

$$\Delta_n = \begin{vmatrix} a_1 & a_3 & a_5 & \cdots & 0 & 0 & 0 \\ a_0 & a_2 & a_4 & \cdots & \cdot & \cdot & \cdot \\ 0 & a_1 & a_3 & \cdots & a_n & 0 & 0 \\ 0 & a_0 & a_2 & \cdots & a_{n-1} & 0 & 0 \\ \cdot & \cdot & \cdot & & a_{n-2} & a_n & 0 \\ \cdot & \cdot & \cdot & & a_{n-3} & a_{n-1} & 0 \\ 0 & 0 & 0 & \cdots & a_{n-4} & a_{n-2} & a_n \end{vmatrix}$$

where we substituted zero for a_s if $s > n$. For all the roots to have negative real parts, it is necessary and sufficient that successive principal minors of Δ_n be positive. The successive principal minors are the following determinants:

$$\Delta_i = \begin{vmatrix} a_1 & a_3 & \cdots & a_{2i-1} \\ a_0 & a_2 & \cdots & a_{2i-2} \\ 0 & a_1 & \cdots & a_{2i-3} \\ \cdot & \cdot & & \cdot \\ 0 & 0 & \cdots & a_i \end{vmatrix} \qquad (i = 1, 2, \ldots, n - 1)$$

where $a_s = 0$ if $s > n$. (It is noted that some of the conditions for the lower-order determinants are included in the conditions for the higher-order determinants.) If all these determinants are positive, and $a_0 > 0$ as already assumed, the equilibrium state of the system whose characteristic equation is given by Equation (5–37) is asymptotically stable. Note that exact values of determinants are not needed; instead, only signs of these determinants are needed for the stability criterion.

Now consider the following characteristic equation:

$$a_0 s^4 + a_1 s^3 + a_2 s^2 + a_3 s + a_4 = 0$$

Obtain the condition for stability using the Hurwitz stability criterion.

Solution. The conditions for stability are that all the a's be positive and that

$$\Delta_2 = \begin{vmatrix} a_1 & a_3 \\ a_0 & a_2 \end{vmatrix} = a_1 a_2 - a_0 a_3 > 0$$

$$\begin{aligned} \Delta_3 &= \begin{vmatrix} a_1 & a_3 & 0 \\ a_0 & a_2 & a_4 \\ 0 & a_1 & a_3 \end{vmatrix} \\ &= a_1(a_2 a_3 - a_1 a_4) - a_0 a_3^2 \\ &= a_3(a_1 a_2 - a_0 a_3) - a_1^2 a_4 > 0 \end{aligned}$$

It is clear that, if all the a's are positive and if the condition $\Delta_3 > 0$ is satisfied, the condition $\Delta_2 > 0$ is also satisfied. Therefore, for all the roots of the given characteristic equation to have negative real parts, it is necessary and sufficient that all the coefficients a's are positive and $\Delta_3 > 0$.

A-5-10. Show that the Routh's stability criterion and Hurwitz stability criterion are equivalent.

Solution. If we write Hurwitz determinants in the triangular form

$$\Delta_i = \begin{vmatrix} a_{11} & & & & * \\ & a_{22} & & & \\ & & \cdot & & \\ & & & \cdot & \\ & & & & \cdot \\ 0 & & & & a_{ii} \end{vmatrix} \qquad (i = 1, 2, \ldots, n)$$

where the elements below the diagonal line are all zeros and the elements above the diagonal line any numbers, then the Hurwitz conditions for asymptotic stability become

$$\Delta_i = a_{11}a_{22}\cdots a_{ii} > 0 \qquad (i = 1, 2, \ldots, n)$$

which are equivalent to the conditions

$$a_{11} > 0, \qquad a_{22} > 0, \qquad \ldots, \qquad a_{nn} > 0$$

We shall show that these conditions are equivalent to

$$a_1 > 0, \qquad b_1 > 0, \qquad c_1 > 0, \qquad \ldots$$

where $a_1, b_1, c_1, \ldots$ are the elements of the first column in the Routh array.

Consider, for example, the following Hurwitz determinant, which corresponds to $n = 4$:

$$\Delta_4 = \begin{vmatrix} a_1 & a_3 & a_5 & a_7 \\ a_0 & a_2 & a_4 & a_6 \\ 0 & a_1 & a_3 & a_5 \\ 0 & a_0 & a_2 & a_4 \end{vmatrix}$$

The determinant is unchanged if we subtract from the ith row k times the jth row. By subtracting from the second row a_0/a_1 times the first row, we obtain

$$\Delta_4 = \begin{vmatrix} a_{11} & a_3 & a_5 & a_7 \\ 0 & a_{22} & a_{23} & a_{24} \\ 0 & a_1 & a_3 & a_5 \\ 0 & a_0 & a_2 & a_4 \end{vmatrix}$$

where

$$a_{11} = a_1$$

$$a_{22} = a_2 - \frac{a_0}{a_1}a_3$$

$$a_{23} = a_4 - \frac{a_0}{a_1}a_5$$

$$a_{24} = a_6 - \frac{a_0}{a_1}a_7$$

Similarly, subtracting from the fourth row a_0/a_1 times the third row yields

$$\Delta_4 = \begin{vmatrix} a_{11} & a_3 & a_5 & a_7 \\ 0 & a_{22} & a_{23} & a_{24} \\ 0 & a_1 & a_3 & a_5 \\ 0 & 0 & \hat{a}_{43} & \hat{a}_{44} \end{vmatrix}$$

where

$$\hat{a}_{43} = a_2 - \frac{a_0}{a_1} a_3$$

$$\hat{a}_{44} = a_4 - \frac{a_0}{a_1} a_5$$

Next, subtracting from the third row a_1/a_{22} times the second row yields

$$\Delta_4 = \begin{vmatrix} a_{11} & a_3 & a_5 & a_7 \\ 0 & a_{22} & a_{23} & a_{24} \\ 0 & 0 & a_{33} & a_{34} \\ 0 & 0 & \hat{a}_{43} & \hat{a}_{44} \end{vmatrix}$$

where

$$a_{33} = a_3 - \frac{a_1}{a_{22}} a_{23}$$

$$a_{34} = a_5 - \frac{a_1}{a_{22}} a_{24}$$

Finally, subtracting from the last row $\hat{a}_{43}/a_{33}$ times the third row yields

$$\Delta_4 = \begin{vmatrix} a_{11} & a_3 & a_5 & a_7 \\ 0 & a_{22} & a_{23} & a_{24} \\ 0 & 0 & a_{33} & a_{34} \\ 0 & 0 & 0 & a_{44} \end{vmatrix}$$

where

$$a_{44} = \hat{a}_{44} - \frac{\hat{a}_{43}}{a_{33}} a_{34}$$

From this analysis, we see that

$$\Delta_4 = a_{11}a_{22}a_{33}a_{44}$$

$$\Delta_3 = a_{11}a_{22}a_{33}$$

$$\Delta_2 = a_{11}a_{22}$$

$$\Delta_1 = a_{11}$$

The Hurwitz conditions for asymptotic stability

$$\Delta_1 > 0, \qquad \Delta_2 > 0, \qquad \Delta_3 > 0, \qquad \Delta_4 > 0, \qquad \ldots$$

reduce to the conditions

$$a_{11} > 0, \qquad a_{22} > 0, \qquad a_{33} > 0, \qquad a_{44} > 0, \qquad \ldots$$

The Routh array for the polynomial

$$a_0 s^4 + a_1 s^3 + a_2 s^2 + a_3 s + a_4 = 0$$

where $a_0 > 0$, is given by

$$\begin{matrix} a_0 & a_2 & a_4 \\ a_1 & a_3 & \\ b_1 & b_2 & \\ c_1 & & \\ d_1 & & \end{matrix}$$

From this Routh array, we see that

$$a_{11} = a_1$$

$$a_{22} = a_2 - \frac{a_0}{a_1} a_3 = b_1$$

$$a_{33} = a_3 - \frac{a_1}{a_{22}} a_{23} = \frac{a_3 b_1 - a_1 b_2}{b_1} = c_1$$

$$a_{44} = \hat{a}_{44} - \frac{\hat{a}_{43}}{a_{33}} a_{34} = \frac{b_2 c_1 - b_1 c_2}{c_1} = d_1$$

Hence the Hurwitz conditions for asymptotic stability become

$$a_1 > 0, \qquad b_1 > 0, \qquad c_1 > 0, \qquad d_1 > 0, \qquad \ldots$$

Thus we have demonstrated that Hurwitz conditions for asymptotic stability can be reduced to Routh's conditions for asymptotic stability. The same argument can be extended to Hurwitz determinants of any order, and the equivalence of Routh's stability criterion and Hurwitz stability criterion can be established.

A-5-11. Show that the first column of the Routh array of

$$s^n + a_1 s^{n-1} + a_2 s^{n-2} + \cdots + a_{n-1} s + a_n = 0$$

is given by

$$1, \qquad \Delta_1, \qquad \frac{\Delta_2}{\Delta_1}, \qquad \frac{\Delta_3}{\Delta_2}, \qquad \ldots, \qquad \frac{\Delta_n}{\Delta_{n-1}}$$

where

$$\Delta_r = \begin{vmatrix} a_1 & 1 & 0 & 0 & \cdots & 0 \\ a_3 & a_2 & a_1 & 1 & \cdots & 0 \\ a_5 & a_4 & a_3 & a_2 & \cdots & 0 \\ \cdot & \cdot & \cdot & \cdot & & \cdot \\ \cdot & \cdot & \cdot & \cdot & & \cdot \\ \cdot & \cdot & \cdot & \cdot & & \cdot \\ a_{2r-1} & \cdot & \cdot & \cdot & \cdots & a_r \end{vmatrix}$$

$$a_k = 0 \qquad \text{if } k > n$$

Solution. The Routh array of coefficients has the form